

Which Escalation Signal Pays Depends on the Traffic: Free Baselines, Corpus Ceilings and Graded Actions in Sub-Billion VLMs

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github.com/thekester/gwel

Abstract

Vision-language cascades answer from a downsampled image and escalate to full resolution when the model appears uncertain. Published policies learn that decision by reinforcement learning on 7B-class models and read it from the output distribution after generation. We ask what a 500M model can supply before it writes a token, and price every candidate against the baseline this literature omits: randomising between fixed configurations, which needs no signal and traces a convex hull any adaptive policy must clear.

The answer inverts between two regimes. Across a four-dataset mixture, output entropy never clears the hull, because reading it requires completing the pass it prices; a probe on pre-generation activations clears it, but so do the image’s own pixel dimensions, read from the file header for nothing and within 0.010 of the probe at every operator preference, and a policy given only the dataset label dominates both. Inside a workload the ranking reverses: image size falls to 0.508 AUROC and clears nothing on either of the two workloads measured, while output entropy on a graded resolution ladder clears by +0.041 to +0.001 on the one where the model is strong and at one preference of four on the one where it is not, and binary escalation never clears at all. **Which signal is worth reading is a property of the traffic, and a method evaluated only on a mixture will recommend the wrong one for a workload.** The probe’s apparent advantage (0.760 against 0.617) is a between-domain effect, and it is one of eight claims lost to family-wise correction over the sixty-three paired comparisons reported here.

What survives is graded action and measured cost. On 1200 document pages, 93% of what escalation repairs is repaired below the top rung, and accuracy saturates at a pixel resolution shared by four models spanning $8.6\times$ parameters, two lineages and three tokenisers, and by a second corpus that reaches the same target while spending half the visual tokens to get there. A fifth model, whose tile grid moves independently of its input, makes that ceiling an upper bound rather than a constant and separates the quantities the others confound: one pixel rung at a fixed token budget buys +0.108, while tripling the token budget at fixed pixels buys +0.024 [−0.006, +0.054]. We state the procedure that locates the ceiling and the sample size it needs.

1 Introduction

A vision-language model asked “what does the sign say?” can answer from a thumbnail or from the full-resolution image. The thumbnail costs a fraction as much and is often sufficient. Deciding *per query* which to use is the premise of adaptive visual perception, and the raw material for it is there: on our pilot, 34% of queries are answered correctly at 384px and would gain nothing from more pixels, 41% fail at both resolutions, and escalation changes the answer on 21%. Whether a per-query decision *pays*, against the simpler move of randomly assigning queries to fixed configurations at the same expected cost, is a separate question, and one this paper ends up answering more narrowly than it expected to.

Published systems make this decision by learning it and by reading it after generating: VisionThink [1], AdaptVision [2] and AwaRes [3] all train an escalation policy with reinforcement learning on 7B-class models, supervised by judge models of their own.

Below one billion parameters none of that is available. Curating twenty thousand samples and running GRPO is not an option for a model that exists because it fits in a gigabyte of GPU memory [9]. Our question is therefore narrow: *what does a 500M VLM already know about its own need for more pixels, and how cheaply can that be read?*

Fig. 1 summarises the serving path: a single cheap pass, a score, and a choice among resolution rungs priced by measured latency. The answer depends on the evaluation distribution. On a benchmark mixture the model appears to know more than its output reveals; inside a single workload most of that apparent knowledge is a recognition of which benchmark the query came from. What survives is narrower and, we think, more useful: reading the right quantity matters more than reading it well, deciding *how far* to escalate matters more than deciding *whether*, and what the model knows is whether to look again rather than where.

Contributions.

1. **An evaluation protocol for allocation signals, and the failure it detects.** Pooling examples across benchmarks that differ in how much resolution buys inflates any signal correlated with benchmark identity. Applied to a pre-generation probe, per-example cost accounting exposes it escalating 1.19 to $1.33\times$ larger

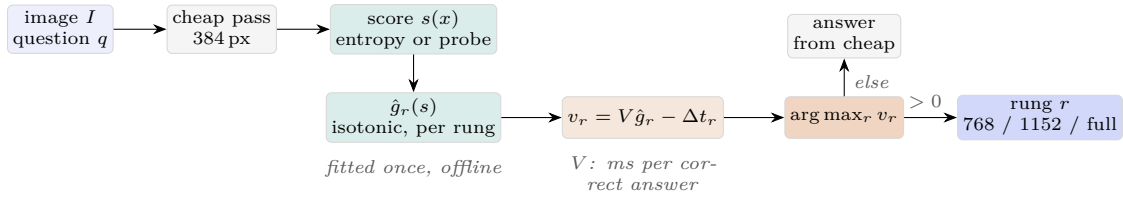


Figure 1: The serving path. Everything up to the score is a single cheap pass; the calibrators are fitted once offline and the operator supplies one number, V . Which rung is chosen, and whether any is, falls out of the measured latencies rather than a tuned rate (Algorithm 2).

images than entropy; scored inside a single dataset it reaches 0.572 where output entropy, fitted on nothing, reaches 0.663, at every depth and for four probe families; and the disagreement between the two signals predicts the source dataset at 0.738. The ranking advantage is a between-domain effect and does not survive family-wise correction (§6, §6.1).

2. Which signal pays is a property of the traffic, and the two regimes require opposite answers.

Measured against randomisation between fixed configurations, which needs no signal and which the escalation literature does not use as a baseline: across a benchmark mixture, raw image size read from the file header clears the hull at every operating point and stays within 0.010 of a probe on activations, while output entropy never clears it; inside one workload the ranking inverts, image size falls to 0.508 AUROC and clears nothing, and only entropy on a graded action space clears the hull (+0.041 to +0.001). A policy that randomises with the probe’s mid-prefill abort clears nothing either, which places the probe’s margin in its signal and not in its read cost. A method evaluated only on a mixture will recommend the wrong signal for a workload (§5.1, §7.2).

3. A decision rule derived from measured cost, and the cascade assumption it corrects.

Calibrating the expected signed gain and thresholding at the cost-implied break-even removes the rate parameter. Calibrating the weak model’s error probability instead, the standard cascade recipe, coincides with our rule exactly when escalation cannot make things worse, and spends 59% relatively more compute here because it can (§7). The rule is a property of the method, not of any signal: run on output entropy inside each domain, one unchanged operator preference yields escalation rates from 1% to 61% tracking what escalation is worth there ($r = 0.974$) (§7.1).

4. How far to escalate, and where it stops paying.

Of the queries escalation repairs, 77% are repaired by an intermediate rung, and that rung is twice as efficient per millisecond as the one above it. On a single-domain pilot with rungs verified distinct per example, accuracy saturates at a pixel resolution that four models agree on, spanning $8.6\times$ parameters,

two lineages and three ways of turning pixels into tokens, and by a second corpus that reaches it while spending half the visual tokens, so a token budget cannot be what runs out. The ceiling is measurable once by a procedure we state and price at about 300 pages. A fifth model, whose token budget stays flat as resolution rises, saturates a rung earlier, which makes the corpus ceiling an upper bound rather than a constant (§7.2, §6.1).

5. A measured cost model. Every comparison here is a cost comparison, so we measure rather than assume, and four standard proxies mispredict: the vision encoder is 43 to 50% of the visual cost, which caps post-encoder pruning at half; decode latency follows depth rather than parameter count; a flat per-configuration price understates escalation by 16%; and for 56% of the mixture the “full-resolution” configuration is not a distinct configuration at all (§3.2, §7.2).

How to read this paper. Three questions are asked in sequence and they receive different answers. §5 to §6.1 ask what a pre-generation signal is worth and conclude that on heterogeneous traffic it is worth what the image dimensions are worth, which is to say it need not be read from the model at all. §7 asks how a threshold should be set once a signal is chosen, and removes the rate parameter given a stated preference. §7.2 asks how far to escalate, and finds the answer to be a property of the corpus rather than of the query, measurable once by a stated procedure. A reader interested only in deployment can read §7.2 and Algorithm 3 alone.

Beyond these, the paper reports several bounded results in their sections: the untrained self-report mechanism loses to uncertainty routing (§5.1); a localizer on pooled visual-token states beats random cell choice by 0.9 points, 6% of the gap a perfect one would close (§A); an ablation read at fixed budget rather than fixed preference (§B); and abstention priced on both of its axes (§E).

2 Related work

Adaptive visual acquisition. The three systems closest to this work all learn when to look closer. VisionThink [1] trains a special token requesting the full-

resolution image with GRPO and an LLM-as-judge reward, plus a penalty calibrated to stop the policy collapsing into always escalating. AdaptVision [2] replaces the binary decision with a bounding-box tool call and introduces Decoupled Turn Policy Optimization to separate the credit assigned to tool tokens from that assigned to answer tokens. AwaRes [3] makes the coupled decision of *whether* and *where* jointly, with cold-start supervised fine-tuning followed by multi-turn GRPO, supervised automatically by a LLaMA-3.3-70B judge and an oracle grounding model. All three evaluate at 7B scale, all three learn the decision rather than reading it, and all three condition on the output of a completed pass. We differ on each count, and §5.1 measures what the first of those choices is worth by implementing their mechanism without its training.

Visual token compression. A parallel line reduces cost by discarding tokens after the vision encoder has run, typically ranking them by attention score. GlimpsePrune [4] is the variant we compare against in detail: it learns a data-driven pruning metric from a single-pass “glimpse” and removes 92.6% of visual tokens at parity. These methods and ours act on different halves of the same cost, which §3.2 quantifies: post-encoder pruning cannot recover the encoder’s share, and on the three models we measure that share is 43 to 50%.

Routing from internal states. In the text setting, several groups have shown that a model’s activations predict its own success before it generates. Lugoloobi et al. [5] train linear probes on pre-generation activations and route across a model pool at 40 to 70% lower cost; Moreno Cencerrado et al. [6] isolate a linear “in-advance correctness direction” that saturates at intermediate layers; NVIDIA’s LLM Router [7] predicts a target model’s correctness from a *different* model’s prefill activations, closing 45.6% of its oracle gap. Every probe in this literature predicts whether the model will succeed. None predicts whether an intervention would change the outcome, because for text LLMs the intervention is extended reasoning over an unchanged input and the two questions nearly coincide. For a VLM they do not, and §3.3 shows they come apart empirically.

Where this sits. Reproducing GRPO training at 7B is not affordable here, so we do not claim a numeric comparison against the systems below; what can be stated precisely is what each one reads, when it decides, and what supervision it needs (Table 1).

The distinction that matters for cost is the third column. Three of the four learn a decision the model *emits*, which requires generation to begin; ours and GlimpsePrune’s read a state the forward pass has already computed. We note that GlimpsePrune measures its own overhead end to end and reports prefill at 69.1% of baseline, so the discipline of charging for the decision is not unique to

method	reads	when	trained by
VisionThink [1]	special token	during	GRPO
AdaptVision [2]	tool call	during	DTPO
AwaRes [3]	policy output	during	SFT+GRPO
GlimpsePrune [4]	glimpse token	$\frac{2}{3}L$	a head
entropy baseline	output scores	after	n/a
image size	file header	before	n/a
ours	residual stream	layer ℓ	n/a

Table 1: What each method reads and what it costs to train. “During” means the decision is a generated token, available only once decoding starts; “before” means the decision needs no forward pass at all, which is the baseline §5.1 finds hardest to beat. The three trained policies also need a supervisor: an LLM judge, GPT-4o, and a LLaMA-3.3-70B judge with an oracle grounding model respectively. GlimpsePrune prunes tokens rather than choosing resolution, and is listed because it is the one method that also reads an intermediate layer.

this work, only unusual among the resolution-choosing methods.

Calibrated cascade routing. A separate line takes the routing score as given and asks how the threshold should be set. Kotte [11] argues the threshold should not be tuned at all: calibrate the uncertainty signal to a probability by isotonic regression, and threshold policies on that probability are cost-optimal under three stated assumptions. Ruan et al. [12] set gates by an exact lower bound on the survival rate of successful cases, so the policy carries a recall guarantee and cannot degenerate into never escalating, and Tayebati et al. [13] obtain distribution-free coverage for a three-way answer/hedge/abstain split. §7 adopts the first and shows why its central assumption fails for a visual cascade.

Attacks on the allocation mechanism. Liu et al. [15] learn a universal trigger, confined to an image border, that lowers a weak model’s confidence and so forces deferral to the expensive model: the provider pays and accuracy metrics never register it. Their attack is metric-agnostic because it flattens the output distribution itself, degrading perplexity, token-margin and verbalised confidence alike; Sun et al. [16] report the same failure mode for text cascades. Both target signals read *after* generation, which is where every visual escalation method we have surveyed reads.

Cost measurement. Zhan et al. [8] profile on-device VLM energy and find average power to be a model fingerprint, so energy reduces to time; their headline that decode dominates 86 to 97% of it is measured under prompts eliciting hundreds of output tokens, and their own Table 3 shows that fraction collapsing to 0.04 for short answers. Shin et al. [10] decompose sub-3B VLM latency by component on Jetson hardware and report that theoretical

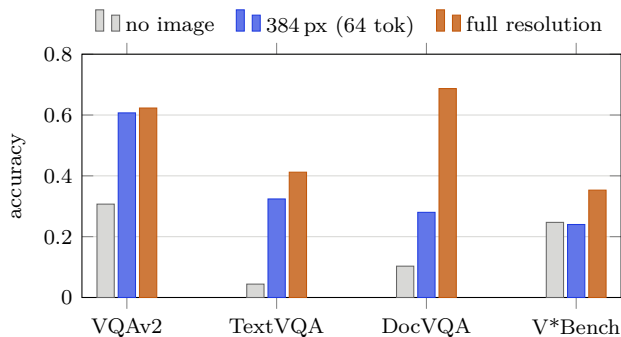


Figure 2: Accuracy per dataset at three information levels, 1000-example pilot. The gap between the blue and orange bars is everything escalation can ever buy, and it ranges from 1.6 points on VQAv2 to 40.7 on DocVQA. On V*Bench the thumbnail matches the blind baseline: at 384 px its small objects are simply gone.

reductions do not predict on-device cost. We adopt their decomposition and reach the same conclusion by a different route in §3.2.

3 Setup

3.1 Configurations and oracle

For each example (I, q, a^*) we run a set $\mathcal{C}$ of visual configurations and record, for every $c \in \mathcal{C}$, the answer, its correctness, the model’s per-step generation scores, and hardware measurements. Configurations comprise a blind baseline, low-resolution previews at two sizes (ANSWER-LOW), a resolution-capped full pass, one preview+crop per cell of a 2×2 grid (CROP), and preview+OCR transcript variants (OCR). The full pilot is 1000 examples across VQAv2, TextVQA, DocVQA and V*Bench, giving 13 000 instrumented generations.

Where resolution matters is a property of the benchmark. Figure 2 is the observation the rest of the paper has to be read against. Cutting the input to 64 visual tokens costs VQAv2 1.6 accuracy points (0.623 to 0.607), and 31% of it is answered correctly with no image at all; DocVQA loses 40.7 points to the same cut, and correctness flips between the two passes on 50% of its queries against 8% for VQAv2. This is the sub-billion transposition of what [1] report at 7B, where a $4\times$ token reduction is nearly free outside OCR-heavy benchmarks. The consequence for routing is immediate: in any mixed corpus the escalation signal is entangled with dataset identity from the start, and §6 measures how much of the probe’s apparent skill is exactly that.

Correctness uses the metric each benchmark defines (VQA accuracy for VQAv2 and TextVQA, ANLS for DocVQA, exact match for V*Bench multiple choice) and is recomputed offline from stored answers, so changing the

metric never requires re-running the model.

The oracle labels each example with the cheapest configuration that answered correctly, under

$$J(c) = w_e \mathbf{1}[\hat{a}_c \neq a^*] + \lambda_t t_c + \lambda_e E_c + \lambda_m M_c + \lambda_v V_c, \quad (1)$$

with t latency, E energy above a measured idle baseline, M peak memory and V visual tokens. Weights in Eq. 1 are calibrated so each resource term contributes 0.03 to 0.09 against an error weight of 1.0: the oracle requires correctness first, then discriminates on measured cost.

What the weights can and cannot change. Calibrating them so each resource term lands in a narrow band is a convention, not an argument, and Proposition 1 minimises this function, so a reader is entitled to ask what a different choice would do. Two answers, one structural and one measured.

Structurally, **oracle accuracy cannot depend on the weights at all.** The oracle ranks only among configurations that answered correctly, so no setting of λ can make a wrong answer preferable to a right one, and the set of solvable examples is fixed before any weight is read. Sweeping every weight over four orders of magnitude confirms it exactly: accuracy stays at 0.694 to within 10^{-9} .

For the same reason w_e **is inert in the labelling:** it is added to options the oracle has already discarded, so 100% of labels survive a $10\,000\times$ change in it. The error weight matters only through the ratio $V = w_e/\lambda_t$ in the serving rule of §7, where it is swept rather than fixed. Our own calibration story, phrased as resource terms measured “against an error weight of 1.0”, therefore describes something the labelling never consults.

What the weights do choose is *which* correct action is cheapest. A hundredfold rise in λ_t moves the oracle’s mix from 16% to 34% CROP and from 11% to 2% OCR, leaving 75% of labels unchanged. That is not a fragility, it is the budget-dependence this paper reports as a finding: when visual tokens are dear, a preview plus a transcript beats a high-resolution crop. The reader should treat every accuracy claim here as weight-free and every action-mix claim as conditional on a stated budget.

Two further properties of Eq. 1 are worth stating. Being linear in the resources, it expresses only finitely many policies: sweeping λ_v over its whole range on our pilot yields exactly two, CROP below a crossover and ANSWER-LOW above. And a policy conditioned on a probe must be charged for the probe, so an escalated query pays both passes throughout this paper.

3.2 What a pass actually costs

Every policy comparison in this paper is a cost comparison, so the cost model has to be measured rather than assumed. We time each stage of one pass with explicit

CUDA synchronisation, following the component decomposition of [10]. Two standard proxies turn out to mispredict, and both matter for what follows.

The vision encoder is half the visual cost. Table 2 attributes latency on three models spanning 8.6× in parameters and two distinct vision encoders.

model	encoder	extra prefill	encoder share
SmolVLM-256M	52.5	55.0	49%
SmolVLM-500M	52.4	53.5	50%
SmolVLM2-2.2B	193.6	252.8	43%

Table 2: Visual cost split at 768 px, in milliseconds.

This bounds a whole family of methods. Post-encoder pruning [4] can only recover the prefill half, because the encoder has already run when tokens are scored; reducing input resolution recovers both. **Token pruning is capped at roughly half the available saving**, and the ratio holds across the parameter range. It also tells us what a probe reading mid-prefill can and cannot save, which §5 makes precise.

Decode latency follows depth, not size. SmolVLM2-2.2B emits the same three-token answer in 47.2 ms against 77.0 ms for the 4.4× smaller 500M model, because it has 24 layers rather than 32; per-layer cost is roughly constant at 2.0 to 2.4 ms. Choosing an edge model by parameter count optimises the wrong variable, and a cost model linear in parameters, as the power fit of [8] is, will not transfer here.

Cache reuse cannot refund the cheap pass. Our accounting charges an escalated query for both passes with no reuse, and the obvious objection is that a serving stack would not. VLCache [17] reports 1.2 to 16× time-to-first-token speedups by reusing encoder and KV caches across multimodal requests. Two facts close this, and neither needs an assumption about what a stack achieves.

First, escalation is a cache miss by construction: VLCache keys its cache on a hash over the input pixels, and escalating submits different pixels. That literature refunds *repetition*, and escalation is the opposite of repetition.

Second, and independently, decode is uncacheable. Reading output entropy requires the cheap pass to *generate*; generation produces the answer, differs per query, and no cache returns it. Writing c for the fraction of encoder and prefill a cache refunds, the probe’s saving per escalated query is $(1 - c)(1 - \ell/L)t_{\text{prefill}} + t_{\text{decode}}$, which runs from 103.1 ms at $c = 0$ to a floor of 64.7 ms at a perfect cache, still 63% of the uncached saving. No refund fraction erases it.

Energy is not reported. Our NVML integration disagrees by 18 to 96% between configurations spending

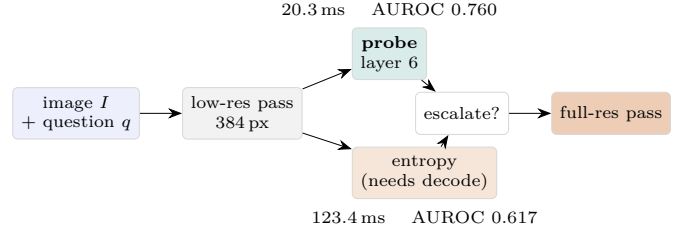


Figure 3: Two allocation signals read from the same low-resolution pass. Output entropy requires the pass to complete and generate; the probe reads the residual stream at layer 6 of 32 and needs no decode.

target	output entropy	pre-generation probe
Q_1 correct	0.758	0.822
Q_2 escalation helps	0.617	0.760
Q_3 answerable	0.539	0.681

Table 3: AUROC by target. Entropy is scored with the sign that favours it; read with the Q_1 sign it scores 0.372 on Q_2 , because the sign flips between the two questions.

identical visual tokens, while their latencies agree within 4 to 5%; a standing check refuses to report joules until that is repaired. This also bears on reading [8], whose headline, decode dominates 86 to 97% of inference energy, is measured under prompts eliciting hundreds of output tokens. Their own Table 3 shows the decode fraction collapsing to 0.04 under short-answer prompts, the regime every dataset here occupies. All costs in this paper are therefore latencies.

3.3 The three routing questions

Let y_{cheap} and y_{full} be the correctness of the low-resolution and full-resolution passes. Three distinct binary targets can be formed over the same examples:

$$Q_1 : y_{\text{cheap}} \quad (\text{is the cheap pass correct?}) \quad (2)$$

$$Q_2 : \neg y_{\text{cheap}} \wedge y_{\text{full}} \quad (\text{does escalation recover?}) \quad (3)$$

$$Q_3 : \bigvee_{c \in \mathcal{C}_{\text{route}}} y_c \quad (\text{answerable at all?}) \quad (4)$$

The escalation literature conditions on Q_1 . The quantity that decides whether compute is well spent is Q_2 , evaluated on the subset where the cheap pass failed. Q_3 separates queries no visual operation can rescue.

4 Confidence answers the wrong question

Mean token entropy over the generated answer predicts Q_1 well and Q_2 poorly. Table 3 reports AUROC on held-out folds, $n = 1000$ overall and $n = 617$ for Q_2 .

The sign flip is the mechanism. A *confident* wrong answer is a knowledge failure that more pixels do not fix;

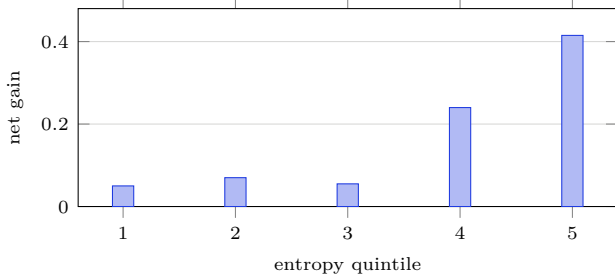


Figure 4: Net benefit of escalating, by entropy quintile ($n = 1000$, 200 per bin). AUROC of entropy for predicting net benefit is 0.734.

an *uncertain* wrong answer is often a perception failure that they do. Splitting the 617 failures at their median entropy, escalation recovers 25% of confident failures and 42% of uncertain ones.

The two effects reinforce for a policy. Writing the net benefit of escalating as

$$G = \mathbf{1}[\neg y_{\text{cheap}} \wedge y_{\text{full}}] - \mathbf{1}[y_{\text{cheap}} \wedge \neg y_{\text{full}}], \quad (5)$$

$\mathbb{E}[G]$ rises monotonically across entropy quintiles, from +5% to +41.5% (Fig. 4). Escalation is worth eight times more in the top quintile than the bottom.

Escalation is also not monotone: 4.1% of queries are answered correctly at 384 px and wrongly at full resolution. No surveyed method models this risk.

5 A pre-generation probe

Following [5, 6], we read the residual stream $h^{(\ell)} \in \mathbb{R}^d$ at the last prompt position (after the instruction, before any token is generated) and fit the difference-of-means direction

$$w^{(\ell)} = \mu_+^{(\ell)} - \mu_-^{(\ell)}, \quad s(h) = \frac{(h - \bar{\mu}^{(\ell)})^\top w^{(\ell)}}{\|w^{(\ell)}\|}, \quad (6)$$

where $\mu_{\pm}$ are class centroids and $\bar{\mu}$ their midpoint. Eq. 6 is deliberately linear: the question is whether the signal is linearly accessible, not how well a classifier can be tuned. Stated that way it is a framing rather than a defence, so we measured what capacity buys (Table 4).

On the mixture the two-centroid difference *beats* every fitted alternative by 0.048 or more, so the paper’s probe is not an undertrained stand-in for something better; fitting parameters on 700 examples in 960 dimensions costs more than it recovers. And within a domain no family closes the gap to output entropy: the best of the four reaches 0.596 against entropy’s 0.656 [0.645, 0.666]. **The within-domain negative is a property of the representation, not of linear models**, which is the objection this table exists to answer. It remains one family of non-linearities rather than a proof, but it is the family a reviewer would reach for first. Fitting is a single pass

probe family	pooled mixture	within DocVQA
difference of means	0.749	0.576
logistic	0.679	0.570
random Fourier features	0.699	0.596
shrunk centroid (LDA)	0.701	0.579
output entropy	0.735	0.656

Table 4: AUROC on the joint escalation target, 20 resampled folds, layer 6. Random Fourier features give a smooth non-linear boundary without a bespoke architecture; the shrunk centroid is the covariance correction the plain difference of means omits. The within-DocVQA column is fitted on the 1200-page pilot at its full training size; the learning curve of Fig. 9 reports 0.572 for the same probe at 900 training examples over 30 resamples, and Table 11’s 0.519 is the different quantity of fitting inside each of the four mixture datasets on 210 examples apiece.

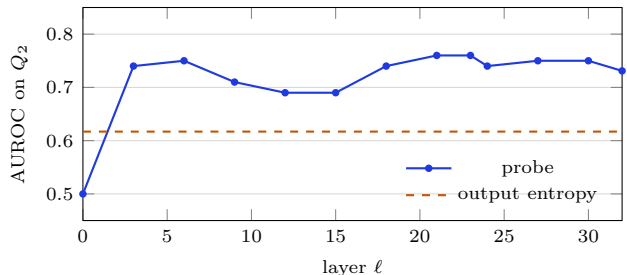


Figure 5: Probe AUROC on Q_2 against depth, single fit per depth on the reported fold; the 0.760 quoted in the text is the bootstrapped estimate at layer 6 over the same fold, which the coarse sweep here renders as 0.750. The ceiling is reached by layer 3 and never improves, so the escalation decision can be made after one sixth of the language model.

over cached activations, and the resulting score is the one consumed by the decision node of Fig. 3.

Depth. Fig. 5 sweeps ℓ . The depth used everywhere in this paper is selected on the validation fold, which independently picks layer 6 (validation AUROC 0.774 on the conditional target, 0.790 on the joint one); the test curve below is illustration, not selection. The signal is absent at $\ell = 0$ by construction, the last prompt token is textual and has attended to nothing, and reaches its ceiling by $\ell = 3$. Bootstrapped, layer 6 gives 0.760 [0.668, 0.849] and layer 23 gives 0.760 [0.652, 0.853]: indistinguishable.

Cost. Decomposing one pass with CUDA synchronisation around each component, a 384 px pass costs 11.0 ms in the vision encoder, 48.1 ms in prefill and 67.8 ms in decode. Reading at layer ℓ needs the encoder plus a fraction of prefill,

$$t_{\text{probe}}(\ell) = t_{\text{enc}} + t_{\text{proj}} + \frac{\ell}{L} t_{\text{prefill}}, \quad (7)$$

policy (escalating 30%)	accuracy	latency	AUROC on G
always cheap	0.385	123.4	n/a
entropy threshold	0.480	182.2	0.751
probe on Q_2	0.395	151.7	0.259
probe on G	0.510	154.3	0.761
always full	0.530	206.0	n/a
oracle	0.580	139.6	n/a

Table 5: End-to-end policy comparison on the held-out fold, latency in milliseconds. Against the entropy threshold, the joint-target probe differs in accuracy by $+0.030$ $[-0.005, 0.070]$, indistinguishable, and in latency by -27.8 $[-40.7, -15.0]$ ms, a significant 15% saving.

giving 20.3 ms at $\ell = 6$ against 123.4 ms for a complete pass.

That ratio is *not* the saving, and it is worth being explicit about why. Entropy requires the cheap pass to finish, but so does answering, whenever the policy decides *not* to escalate. The probe therefore saves only on escalated queries, where the remainder of the cheap pass is abandoned. With escalation rate ρ and full-resolution cost t_{full} ,

$$C_{\text{entropy}} = t_{\text{cheap}} + \rho t_{\text{full}}, \quad (8)$$

$$C_{\text{probe}} = (1 - \rho) t_{\text{cheap}} + \rho (t_{\text{probe}} + t_{\text{full}}). \quad (9)$$

At our observed $\rho \approx 0.30$, Eq. 9 gives 185.3 ms against 154.3 ms: a 17% saving, rising to 31% if every query escalates and falling to 7% at $\rho = 0.10$. The probe is cheaper, but by a sixth, not by the $6\times$ that comparing t_{probe} from Eq. 7 against a full pass would suggest. We state this explicitly because the read-cost ratio is the number an efficiency paper is tempted to report.

5.1 A better ranking is not a better policy

AUROC says a signal orders queries well; it does not say a policy built on it serves better answers per unit of compute. Building both policies at matched escalation rates and charging them by Eq. 9 exposes a trap.

A probe trained on Q_2 (the *conditional* target, defined only on queries the cheap pass failed) produces a policy that is 8.5 accuracy points *worse* than the entropy threshold it beat on AUROC (-0.085 $[-0.130, -0.040]$, paired). The reason is that a deployed policy must decide on every query, not only on failures, and the quantity it needs is the joint event G : escalation helps only when the cheap pass would fail *and* full resolution would succeed. Entropy predicts G at 0.751 because both factors rise with it. The Q_2 -trained probe predicts G at 0.259, far below chance, because it never saw a successful cheap pass.

Retraining the same probe on G fixes it (Table 5).

And the Q_2 row is the more useful result: a probe can beat a baseline on a conditional target and lose decisively as a policy. Papers that report ranking quality without building the policy risk exactly this.

policy	escalates	accuracy	latency
always cheap	0%	0.385	123.4
self-report, “ANSWER or ZOOM”	7%	0.385	137.9
self-report, “detailed enough?”	22%	0.405	169.8
entropy threshold	20%	0.465	164.7
probe	20%	0.475	144.0
always full	100%	0.530	206.0

Table 6: Untrained self-report against uncertainty-based routing, 200 held-out examples, latency in milliseconds. Against the stronger self-report the probe is $+0.070$ $[0.020, 0.120]$ more accurate and 25.8 $[11.9, 39.2]$ ms cheaper, both paired and both significant.

Asking the model directly is worse than reading its uncertainty. The escalation mechanism published systems train is a self-report: shown a downsampled image, the model states whether it needs the full-resolution one. We cannot reproduce VisionThink’s GRPO at sub-1B, that is the premise of this work, but we can implement its *mechanism* without the training, and ask what the untrained version is worth. Two phrasings were tried, both eliciting a one-word judgement (Table 6).

The first phrasing escalates 7% of queries and recovers nothing: its accuracy is exactly that of never escalating, so the compute is pure waste. The second escalates 22% and buys two accuracy points, where routing on entropy at the same rate buys eight and routing on the probe buys nine at lower cost. The self-report is not merely weaker; it escalates as often as the alternatives and selects the wrong queries.

This bounds what the untrained mechanism delivers and, read the other way, quantifies what the training in [1, 2, 3] is buying. It is not a comparison against those systems, which would require reproducing their training at a scale where it is not affordable.

The whole frontier, not one operating point. A single escalation rate flatters or damns arbitrarily. Sweeping it (Fig. 6) gives a cleaner statement: on this fold every entropy operating point is dominated. Of the fourteen policies evaluated, six lie on the accuracy-cost Pareto front, four of them probe-based and none entropy-based.

At matched accuracy the saving is 12% at 0.46 and 22% at 0.52. The 0.52 point matters most: the probe reaches 0.525 at 161.5 ms where always running at full resolution reaches 0.530 at 206.0 ms, within half a point of the accuracy ceiling for 78% of its cost, while the entropy policy at the same accuracy costs 206.9 ms, marginally *more* than simply always escalating.

How much of that is the fold? Universal dominance is a statement about six points on 200 test examples, which is not a scale at which a categorical claim should be made. We therefore refit everything downstream of the split (probe direction, thresholds, calibration) on 200 stratified resplits.

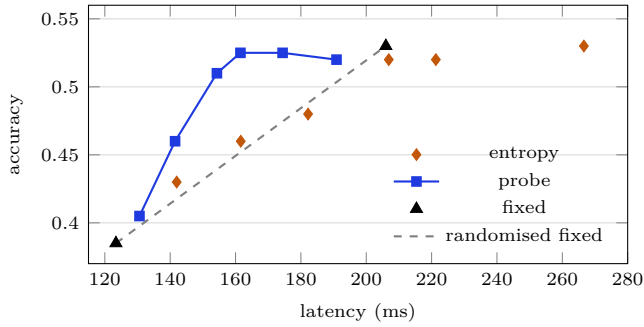


Figure 6: Accuracy against latency as the escalation rate sweeps from 10% to 70%, on the reported fold. The dashed chord is what randomising between the two fixed policies reaches with no signal. On this fold two entropy points sit marginally above it; resampled, the best entropy operating point is indistinguishable from the chord and every other sits below it (Table 7). Four probe points lie on the Pareto front, together with the two fixed policies at its ends. Fig. 7 tests how much of this survives resplitting, and Table 8 adds the two comparators that supersede this comparison: a free image descriptor and a policy that aborts at random.

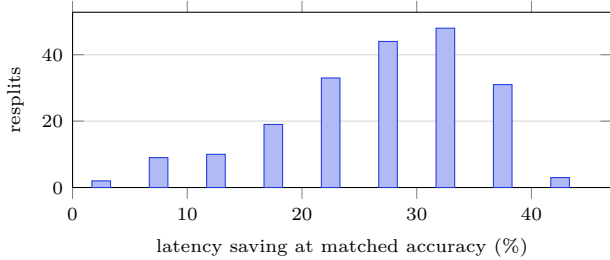


Figure 7: The probe’s latency saving over entropy at matched accuracy, across 200 stratified resplits. The distribution is entirely positive; the single-fold figure of 12 to 22% quoted above is a conservative corner of it.

The universal claim does not survive: all seven entropy operating points are dominated in only 52% of splits. What survives is stronger than that number suggests. On average 0.83 of seven entropy points escape domination, and when one does it wins by a median of 0.010 accuracy, two queries out of two hundred, with 85% of survivors winning by 0.02 or less. Meanwhile the probe holds 90% of the Pareto front, and at matched accuracy it is cheaper in 200 of 200 splits, by a median of 27.8% (Fig. 7).

The honest summary is that the direction of the effect is robust and its universality is not. We report the distribution rather than the fold because the fold is what a single split of 1000 examples can support.

The baseline that needs no signal. Both comparisons above are between adaptive policies, and that is not the bar a deployment sets. Any point on the segment between two fixed policies is reachable by randomisation: serve a

gap to the hull	V=400	800	1600	3200
probe, mixture	+0.054	+0.051	+0.033	+0.008
entropy, mixture	−0.001	−0.016	−0.032	−0.026
ladder, DocVQA	+0.042	+0.029	+0.017	+0.006

Table 7: Accuracy above what randomising between fixed configurations reaches at the same expected latency, 30 resampled folds; every probe and ladder interval excludes zero and every entropy interval excludes any gain above +0.005. Prices here are per configuration; Table 8 repeats the mixture rows under the per-example prices of §6, where the probe’s margin narrows from +0.053 to +0.033 at $V=400$ and the ranking of the policies is unchanged.

fraction of queries at the dear configuration and the rest at the cheap one. The convex hull of the fixed policies is therefore the floor an adaptive policy must clear, and no comparison against a single endpoint establishes it. Table 7 reports each policy’s accuracy minus the hull accuracy at its own expected latency, paired over 30 resampled folds.

The comparison is not an invention of ours: random allocation is a standard baseline in text routing, where [5] plot their cascades against it, and it is absent from the visual escalation evaluations we survey. Nor does the verdict depend on how the operating point is chosen. Sweeping quantile rates instead of preferences, entropy’s best point is indistinguishable from the chord (+0.002 [−0.002, +0.005] at a 10% rate) and every other rate sits significantly below it, while the probe clears the chord at all seven rates (+0.006 to +0.056).

Three readings, each load-bearing. The entropy threshold, the baseline this literature measures against, does not clear the hull here at any operating point of either parameterisation: reading it requires completing the pass it prices, and the read cost eats the routing gain. The probe clears it at every tight and moderate budget, by most where the budget binds, and only at the loosest preference under per-example prices does its interval span zero (−0.001 [−0.004, +0.002] at $V=3200$). Table 8 locates that margin: not in the mid-prefill abort, since randomising with the same abort clears nothing, but in the signal, which §6 shows is largely a domain effect and which a free image descriptor supplies as well. And on the homogeneous pilot the ladder clears the hull where binary escalation does not (§7.2): the graded action space, not the signal, beats the coin flip there. Adaptive escalation pays where its read is cheap or its actions are graded, and not otherwise.

Two comparators that are cheaper still. A hull comparison answers whether a signal beats no signal. It does not answer which part of an adaptive policy is doing the work, and two candidates cost less than the probe.

The first is the free one. §6 finds raw image size scoring 0.748 against the probe’s 0.761, and a signal read from the file header does not merely rank almost as well: it prices

gap to the hull	V=400	800	1600	3200
<i>per-configuration prices</i>				
probe	+0.053	+0.050	+0.033	+0.001
image size (free)	+0.061	+0.050	+0.024	+0.009
entropy	-0.001	-0.012	-0.028	-0.022
random + abort	-0.0005	-0.007	-0.001	-0.001
<i>per-example prices</i>				
probe	+0.033	+0.032	+0.021	-0.001
image size (free)	+0.034	+0.027	+0.011	+0.002
entropy	-0.001	-0.002	-0.014	-0.017
random + abort	-0.0001	-0.006	-0.001	-0.001

Table 8: Accuracy above the randomisation hull at matched expected latency, 30 resampled folds, under both costings. Every image-size interval excludes zero; no random-plus-abort interval does. The probe rows are recomputed on resamples independent of those behind Table 7 and agree with it to within 0.007.

differently. A policy reading it decides before anything runs, so an escalated query never pays for a cheap pass it discards, and its cost structure is exactly that of the randomisation it is being compared against. The second isolates the mechanism rather than the signal: escalate on a coin flip, but abandon the cheap pass mid-prefill as the probe does. If the probe’s margin is owed to the abort, this should reproduce it (Table 8).

Both answers are inconvenient for the apparatus this paper built. **A signal costing nothing to read clears the hull at every operating point of either costing**, and every one of those clearances survives family-wise correction. Paired against the probe over the same resamples, the free signal differs by at most 0.010 either way, and of the preferences where one of them nominally wins, none survives correction. On this mixture the pre-generation probe buys nothing that the image dimensions do not already supply, and the cases where it appears to are the cases the correction removes.

And the abort mechanism, alone, buys nothing at all. Random escalation with the same mid-prefill abandonment sits at or below the hull everywhere, from -0.0005 to -0.007 swept by preference and to -0.022 swept by rate. The margin the probe opens is therefore bought by its signal and not by its read cost, which corrects an attribution an earlier version of this section made in the opposite direction. What the signal knows, given §6, is which dataset the query came from, and image size knows the same thing for free.

And an upper bound on what the confound can be worth. If the signals above are reading dataset identity, the natural ceiling is a policy handed that identity outright. We give one the label and let it pick a rate per dataset, escalating a uniformly random subset inside each. Selection within a dataset stays random by construction, so the policy knows *which benchmark* and nothing

gap to the hull	V=400	800	1600	3200
domain label, tuned rate	+0.080	+0.075	+0.060	+0.051
probe	+0.058	+0.056	+0.034	+0.005
per-domain entropy rule	+0.012	+0.020	-0.022	-0.014

Table 9: What dataset identity alone buys, 30 resampled folds. The first row escalates a random subset within each dataset at a rate tuned per dataset on the test fold, so it is an optimistic bound rather than a deployable policy. The probe row is recomputed on these resamples and agrees with Table 8 to within 0.005.

whatever about the query (Table 9).

The bound dominates every signal in this paper at every preference, while using no information about any individual query. What it does with the label is blunt, and that is the point: unable to choose *which* queries, it escalates almost the whole of the domain where escalation pays and almost none of the domain where it does not, 100% of DocVQA against 13% of VQAv2 averaged over preferences, with V*Bench and TextVQA between them. **On a benchmark mixture, essentially all of the adaptive gain available is the gain from knowing which benchmark you are serving**, and a router that already knows its traffic composition has it for free. That is the sharpest statement of the confound: not merely that the probe correlates with domain, but that domain is the whole of what any of these signals is worth here.

The last row is the complement. Applying one unchanged preference inside each dataset, on output entropy, clears the hull at tight budgets and falls below it at loose ones. Within a domain, once identity is removed, what remains for a per-query signal to earn is small and inconsistently signed, which is §6.1 restated in policy terms rather than in AUROC.

The free descriptor does not survive leaving the mixture. If image size routed inside a single workload too, nothing in this paper would justify reading the model at all. It does not. On the same 1200 pages, the same verified rungs and the same per-example prices, image size predicts the top-rung gain at 0.508, which is chance, against output entropy’s 0.667; and built into a policy it never clears the hull, sitting at or below it at every preference (Table 10).

The two regimes require opposite answers, and this is the paper’s central claim. Across a mixture of benchmarks, escalation value is carried by which benchmark a query came from, a free image descriptor recovers it, and neither output entropy nor a probe on activations earns its read cost. Inside a single workload that between-domain axis is gone: the free descriptor collapses to chance, and what clears the hull is a signal read from the model, spent on a graded action space rather than on a binary one. Neither regime is the general case, and a method evaluated only in the first will recommend the

gap to the hull, DocVQA	V=400	800	1600	3200
ladder, entropy	+0.041	+0.025	+0.010	+0.001
ladder, image size	+0.000	-0.001	-0.004	-0.004
binary, entropy	-0.003	-0.081	-0.017	-0.015
binary, image size	-0.026	-0.052	-0.012	-0.010

Table 10: The same comparison as Table 8, run inside one workload instead of across four. The hull is built from all four fixed rungs, 30 resampled folds, per-example prices. The entropy ladder row is recomputed on independent resamples and agrees with Table 7 to within 0.007.

wrong thing for the second.

The two halves rest on different amounts of evidence and we separate them here. The negative half, that a free descriptor stops working once the between-domain axis is removed, is measured on both workloads we collected. The positive half, that a model-read signal on a graded action space clears the hull, is measured on DocVQA at three preferences of four and on InfographicVQA at one of four (§7.2); on the latter the serving model answers only 0.368 of the corpus at its ceiling, so most queries are unanswerable at any rung and there is little for any policy to allocate. What we claim generally is the inversion of the *ordering* between the two signals. How much the winning side is worth is corpus-dependent, and on a corpus where the model is weak it is worth little.

That is also the reading under which our own negative results stop being awkward. The probe fails inside a domain (§6.1) and is redundant across one; output entropy fails across a mixture (Table 8) and is the only signal that works inside one. Both failures are the same fact seen from two sides: a signal is worth exactly the variance it explains that the free features do not.

A better ranking, again, and again no better policy. Since our own numbers say the ceiling is the signal, we tried to improve the signal. Entropy and the probe are complementary: a logistic combination of the two reaches 0.805 on the joint target, against 0.751 and 0.761 alone, a gain of +0.054 [-0.010, +0.120].

As a policy it is strictly worse than the probe alone (0.505 accuracy at 181.1 ms, against 0.510 at 154.3 ms) because reading entropy requires finishing the cheap pass, which is precisely the cost the probe existed to avoid. The best ranking and the best policy are different objects here, and the second time we observed this it was not a mistake but a structural feature: a signal’s value to a policy is bounded by what it costs to obtain.

How far from the oracle? Not close, and the shape of the gap is itself informative. The oracle escalates only 20% of traffic and is simultaneously more accurate *and* cheaper than always running at full resolution (0.580 at 139.6 ms against 0.530 at 206.0 ms). Measured against the always-cheap baseline, the joint-target probe closes 64%

of the accuracy gap but *increases* cost, where the oracle decreases it. No policy we build moves along the direction the oracle occupies.

This is the strongest evidence in the paper that the ceiling is the signal rather than the policy, and it agrees with what the routing literature reports for text: NVIDIA’s prefill-activation router closes 45.6% of its oracle gap [7], and [5] conclude that routing effectiveness is limited by the reliability of the success estimates, not by the routing rule.

The direction is not domain-general. Training on detail-limited datasets (DocVQA, V*Bench) and testing on knowledge-limited ones (VQAv2) gives 0.422; the reverse gives 0.366. Both fall *below* chance: the direction inverts rather than degrading. The 0.760 result holds only because the training fold spans all four datasets. “Would more pixels help” is not one internal quantity but at least two (one for answers present but unreadable, one for answers absent) represented along opposing axes.

6 What the probe is actually reading

A cost measurement, not a literature review, exposed this. Pricing escalation per example rather than per configuration showed the probe systematically escalating *larger* images than entropy does, 1.19 to 1.33× the visual tokens at matched rate. That is worth explaining, and the explanation undermines the result above.

The flat escalation price was wrong. Our profiling image had a longest side below 1536 px, so the processor capped it and the “full-resolution” pass spent 320 visual tokens. On the pilot, CROP-free full passes average 424 tokens and exceed the 768 configuration on 44% of examples. Latency is affine in token count ($100.9 + 0.326 v$ ms, worst residual 1.7 ms over four profiled configurations) so the honest per-example price is 238.9 ms against the 206.0 we charged, 16% higher. This is the third measurement error of the same family we have found, after processor upscaling and patch-grid quantisation, and all three come from a configuration name not determining a cost.

Under honest costing the probe’s saving over entropy shrinks from -27.8 $[-40.7, -15.0]$ ms to -11.8 $[-27.2, +5.0]$ ms at a 30% escalation rate: no longer distinguishable from zero. It survives at 20% and 40%.

Why a better signal picks dearer escalations. The queries more pixels help are the queries with more pixels to add. A signal that ranks escalation value therefore also selects the expensive tail, and ranking quality partly cancels against cost saving. This is a fourth way a better signal fails a policy, and unlike the first three it is invisible unless cost is measured per query rather than per configuration.

signal	pooled	within one dataset
image size (free)	0.748	0.534
output entropy	0.751	0.664
pre-generation probe	0.761	0.519

Table 11: Predicting the joint escalation gain. Pooled is the held-out fold of the four-dataset mixture, $n = 200$. Within-dataset trains and tests the probe inside a single dataset over 40 resamples, weighted by dataset size, so each fit sees about 210 examples; the single-domain pilot of §6.1 repeats the measurement with nine times the data and reports 0.572 for the probe against 0.663 for entropy.

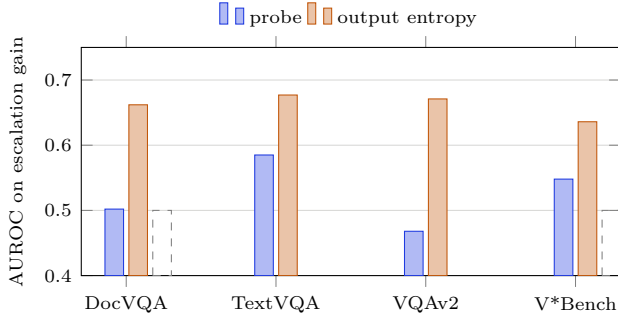


Figure 8: Within a single dataset, trained and tested inside it. The dashed line is chance. The probe’s pooled advantage does not survive removal of the between-domain axis; entropy’s largely does.

And the correlation runs deeper than cost. If image size predicts escalation value, it is a routing signal in its own right, and a free one, read from the file header before any model runs. Table 11 scores it against the two signals the paper has been comparing.

The probe’s margin over a feature costing nothing is $+0.013$. And the within-dataset column is the result (Fig. 8): with the between-domain axis removed the probe is at chance in every dataset (DocVQA 0.502 [0.483, 0.521], VQAv2 0.468 [0.422, 0.513]) while output entropy retains 0.664. Each of these fits sees only 210 training examples, which is a confound of its own; §6.1 removes it with a pilot built for the purpose.

What the disagreement looks like. The statistics above say the probe tracks image size; the queries say what that means. Ranking the held-out fold by each signal and taking the largest rank disagreements, the two sets are almost disjoint by dataset (Table 12).

The disagreement between the two signals predicts which dataset a query came from at AUROC 0.738, which is most of what either signal achieves on the escalation target itself. Where they differ, they differ about what kind of image it is.

The individual queries are unambiguous. The four the probe most wants to escalate are all DocVQA, all 448 to 640 visual tokens at full resolution, and all have $G = 0$:

	DocVQA	TextVQA	VQAv2	V*Bench
test fold	30%	25%	30%	15%
probe favours	50%	36%	12%	2%
entropy favours	0%	28%	46%	26%

Table 12: Dataset composition of the 50 queries where each signal most wants to escalate relative to the other. Not one of the queries entropy prefers is a DocVQA page.

“What is written in big letters on the top right?” is answered *correctly* at 384px and the escalation changes nothing. The four entropy most wants to escalate are all VQAv2 photographs at 320 tokens, all wrong at both resolutions, questions like “What is tucked behind the lamp?” where the model is genuinely lost and more pixels do not help.

Read charitably this is the probe doing something reasonable, since documents do repair more often on average. Read as a routing signal it is a category detector: it escalates a correct answer on a large page and declines a hopeless one on a small photograph, and neither decision used anything about the query.

The text-probe literature saw this coming, and evaluates accordingly. Having found it, we went back to check whether the work we build on is exposed to the same thing. It is not, and one of them names the mechanism. Moreno Cencerrado et al. [6] train and test their correctness direction *within* each dataset and report cross-dataset transfer separately, and observe that for some sources “the learned direction captures dataset-specific cues correlated with correctness rather than correctness itself”. Lugoloobi et al. [5] likewise average AUROC over their benchmarks rather than pooling examples across them. Neither pools; we did.

Two of their observations also validate the test we use here. Cencerrado et al. report that probe performance on GSM8K “plateaus near random chance regardless of dataset size”, distinguishing an absent signal from an under-sampled one by exactly the learning-curve argument, and they find a non-linear classifier on external embeddings generalising *worse* than a linear probe on internal states, the opposite of what a capacity explanation would predict. The correction here is therefore to our evaluation protocol, not to their finding.

This explains a result we had already reported and misread. We found earlier that the probe’s direction inverts across domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) and read it as a transfer limitation calling for a domain-representative calibration set. The correct reading is stronger: a direction that carries no within-domain signal and inverts between domains is a domain classifier, and its pooled AUROC measures how separable the mixture is, not how well the model

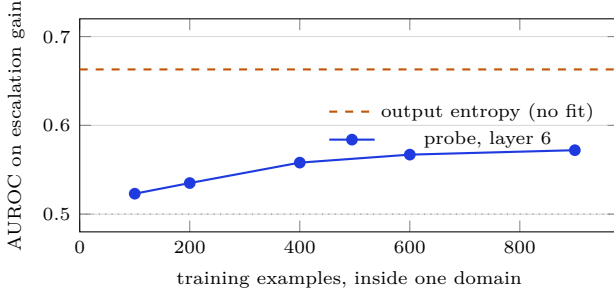


Figure 9: Probe AUROC inside DocVQA against training size, 30 resamples with a fixed 300-example held-out set. The dotted line is chance. More data helps and does not close the gap: at 900 examples, more than the pooled fit that produced 0.761, the probe reaches 0.572.

knows its own needs. Both facts were in our data; only the second explains them.

6.1 A single-domain pilot, and what it settles

The within-domain fit above gets 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. We said so, and then collected the pilot that decides it: 1200 DocVQA pages, enough to fit and test inside one domain.

The decisive form is a learning curve. If the probe were merely starved, AUROC would climb toward the pooled figure as training data grows. Neither extreme is what happens (Fig. 9).

The probe does improve, by +0.050 as training data grows ninefold, so the collapse is not purely starvation. But it plateaus far below output entropy, which is fitted on nothing. **The signal is genuinely weaker inside a domain than the pooled figure suggests**, and the confound conclusion stands in this corrected form rather than as “at chance”.

And the depth that works does not transfer either. We selected layer 6 because AUROC saturates there on the mixture, and the entire read-cost argument of §5 rests on stopping a sixth of the way through the language model. Inside one domain the signal keeps improving with depth:

layer	1	3	6	12	20	32
AUROC	0.531	0.546	0.572	0.579	0.618	0.625

The best depth is the last one. That costs the probe its cheap read: Eq. 7 gives 20.0 ms at $\ell = 6$ and 59.1 ms at $\ell = 32$, so the saving per escalated query falls from 106.9 to 67.8 ms.

That residual 67.8 ms is exactly the decode time, because a full-prefill probe still skips generation, and it is the same floor the cache analysis of §3.2 arrived at from

an unrelated direction. The probe’s advantage narrows to one mechanism, stated twice: reading the model without making it speak.

An independent corroboration of the corrected depth. GlimpsePrune [4] reads its learned glimpse token at layer $K = \frac{2}{3}L$, chosen as the point where it has “aggregated sufficiently clear importance information” at a favourable overhead trade-off. That is where our within-domain sweep becomes informative (0.618 at layer 20 of 32) and far from the layer 6 the mixture selected. Two groups reading different internal signals for different purposes land on the same region of the network, which is weak evidence but it points the same way as §6.1: the shallow saturation was the artefact, not the deep one.

What survives. The mixture result is real, and mixtures are real deployments: a router serving documents and photographs through one endpoint faces exactly the pooled distribution, and there a free image-size feature is not merely nearly as good but, built into a policy and priced, indistinguishable from the probe within 0.010 at every preference (Table 8). The negative results are untouched, since none of them rests on the probe working. And the paper’s thesis survives inverted: we set out to show that a better signal is not a better router, and the sharpest instance turns out to be our own signal, whose advantage was an artefact of the evaluation mixture.

What does not. The claim that a 500M VLM encodes its need for more pixels in a linearly readable direction. On the evidence here it mostly encodes what kind of image it is looking at, which correlates with that need across a benchmark mixture and much less within one. §6.1 is the test we said this needed, run on a corpus built for it, and it does not rescue the claim: given nine times the training data the direction reaches 0.572 where the model’s own output entropy, fitted on nothing, reaches 0.663. The honest statement is that the direction carries some within-domain signal and less than the cheaper alternative.

7 The escalation rate is not a free parameter

Every policy above picks a rate and thresholds at that quantile. That is a tuning exercise, and nothing connects the chosen rate to Eq. 1, the cost function the oracle is already labelled under. Kotte’s UCCI [11] closes the same gap for text cascades (calibrate the signal to a probability, then let the cost model choose the threshold) and its transposition here is short, but it needs one correction.

The rule. Escalating x changes expected cost by

$$\Delta J(x) = -w_e \mathbb{E}[G \mid x] + \lambda_t \Delta t, \quad (10)$$

Algorithm 1 Calibrated gain-optimal escalation

fit

```

1 read  $h^{(\ell)}$  at the last prompt position,  $\ell = 6$ 
2  $G_i \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^{\text{full}}] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^{\text{full}}]$ 
3  $w \leftarrow \mu_+ - \mu_-$  over  $\{G_i > 0\}$  vs. the rest
4  $s_i \leftarrow (h_i - \bar{\mu})^\top w / \|w\|$ 
5  $\hat{g} \leftarrow 2 \text{isotonic}(s, (G+1)/2) - 1$ 
serve  $x$ , given  $V$ 
6  $\Delta t \leftarrow t_{\text{probe}} + t_{\text{full}} - t_{\text{cheap}}$ 
7 if  $\hat{g}(s(x)) > \Delta t/V$  then abandon the prefill, run full
8 else finish the cheap pass and answer
```

calibrator	AUROC	calib. error	accuracy
isotonic	0.660	0.059	0.736
Platt sigmoid	0.667	0.098	0.713
equal-mass bins	0.656	0.051	0.733

Table 13: Calibrator families on 1200 DocVQA pages, 30 resamples, all thresholded at $\tau = 0.243$. Paired against isotonic, Platt loses 0.022 $[-0.026, -0.019]$ accuracy and equal-mass bins differ by $-0.003 [-0.006, 0.000]$.

so the optimal action is to escalate exactly when

$$\mathbb{E}[G \mid x] > \tau = \frac{\Delta t}{V}, \quad V = \frac{w_e}{\lambda_t}, \quad (11)$$

where V is the only knob: the latency an operator will spend to buy one additional correct answer, in milliseconds. Following [14], we state the trade-off in the units of the outcome rather than as a weight per millisecond, because the latter is not readable across devices. Estimating $\mathbb{E}[G \mid x]$ from the probe score by isotonic regression (monotone, non-parametric, one pass over cached activations) gives Algorithm 1.

Why isotonic, and what actually matters about it.

We take isotonic regression from [11], and borrowing a component is how a paper inherits someone else’s assumptions. The settings differ where calibration is sensitive: their target is a probability with tens of thousands of calibration points, ours a signed gain with a few hundred. Three families thresholded at the *same* break-even, so any difference is the calibrator’s magnitudes rather than a different operating point (Table 13).

The sigmoid ranks marginally *best* and calibrates worst, at nearly twice the error, and its policy is significantly less accurate. That is the paper’s recurring pattern arriving in a third place: ranking quality does not transfer, this time because a rule that thresholds a magnitude is broken by a miscalibrated magnitude in a way a rank threshold would never notice. A parametric squash reports gains the operator did not ask for, and the rule lands somewhere other than the point V specifies.

Equal-mass bins, meanwhile, are indistinguishable from isotonic. **The load-bearing property is being non-**

parametric, not being isotonic, and we say so rather than defend a component we happened to inherit. Isotonic remains the better default because monotonicity is free information that binning discards, but nothing here rests on it.

When the rule is optimal, and how it relates to UCCI’s theorem. Proposition 1 below is a direct transposition of the optimality argument of [11] to a different calibration target; we state it to fix notation and to make visible exactly which assumption separates the two rules, not as a contribution in itself. Let $s(x)$ be any routing score and let Π_s be the policies measurable with respect to it, which is what a deployed router is: it sees the score, not the outcome.

Proposition 1. *Under the cost of Eq. 1 with $w_e, \lambda_t > 0$, and given the calibrated gain $\hat{g}(s) = \mathbb{E}[G \mid s]$, the policy*

$$\pi^*(x) = \mathbf{1}[\hat{g}(s(x)) > \Delta t/V], \quad V = w_e/\lambda_t,$$

minimises expected cost over Π_s . If $\hat{g}$ is non-decreasing, it is a threshold on s itself.

Proof. Write $C(\pi) = C_0 + \mathbb{E}[\pi(x) \Delta J(x)]$, where C_0 is the cost of answering cheap on every query and ΔJ is Eq. 10. The expectation is linear in π and the constraint set is $\pi(x) \in \{0, 1\}$, so it is minimised pointwise by escalating exactly where $\Delta J(x) < 0$. For $\pi \in \Pi_s$ the tower property gives $\mathbb{E}[\Delta J \mid s] = -w_e \mathbb{E}[G \mid s] + \lambda_t \Delta t$, so the pointwise test becomes $\hat{g}(s) > \lambda_t \Delta t / w_e$. Monotone $\hat{g}$ makes the sublevel set an interval in s . $\square$

Isotonic regression returns a non-decreasing $\hat{g}$ by construction, so Algorithm 1 realises π^* whenever the calibration is accurate, which Fig. 10 measures rather than assumes. And the relation to Kotte’s Theorem 1 [11] is exact: under their assumption that the escalated pass attains a fixed accuracy γ independent of which queries are sent, $\mathbb{E}[G \mid x] = \gamma - (1 - p_{\text{err}}(x))$, which is increasing in the weak model’s error probability, so their rule and ours coincide. **The two rules agree exactly when escalation cannot make things worse, and Table 14 is what their difference costs when it can.**

The break-even is signal-dependent, and that is the probe’s second advantage. Line 6 differences Eq. 9. A policy reading entropy must finish the cheap pass before deciding, so escalation adds the whole full-resolution pass, $\Delta t = 206.0$ ms. A policy reading the probe abandons the cheap pass it never completed, so $\Delta t = 20.3 + 206.0 - 123.4 = 102.9$ ms, exactly half. At the same operator preference V , the probe’s break-even gain is half entropy’s: **a cheaper signal is entitled to escalate on weaker evidence**, and the saving compounds rather than merely lowering the bill for a fixed rate.

Calibrating the wrong probability. UCCI’s Theorem 1 assumes the strong model attains a fixed accuracy on the escalated subset, invariant to which queries are

rule	calibration target	escalates	accuracy	latency
UCCI [11]	$P(\text{cheap wrong})$	54%	0.520	178.5
ours	$\mathbb{E}[G \mid x]$	34%	0.520	157.9

Table 14: Both rules read the same probe score at $V = 800$ ms per correct answer and differ only in what they calibrate. Paired, the accuracy difference is 0.000 $[-0.025, 0.025]$ and the latency difference $-20.6 [-26.8, -14.9]$ ms: the same answers for 59% relatively more compute.

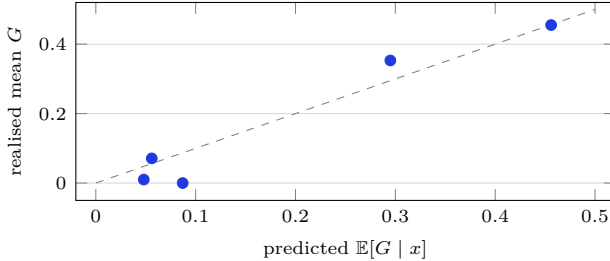


Figure 10: Reliability of the gain calibration on the held-out fold, binned by predicted gain. The dashed line is perfect calibration. A rule that thresholds a *magnitude* is only as good as the magnitude, which a ranking metric would not detect.

sent. Under that assumption the escalation value of a query is determined by the weak model’s error probability, so calibrating $P(\text{cheap wrong})$ suffices. Visual escalation violates it: it is non-monotone, repairing 21% of queries and *damaging* 4%, so the escalated pass is a second random outcome rather than a fixed accuracy. The quantity to calibrate is the signed gain, and the difference is not academic (Table 14).

The failure is legible. A confidently wrong cheap pass has high error probability and near-zero expected gain, so UCCI escalates it and the gain rule does not, the same conditional-versus-joint error of §5.1, displaced from the training target to the calibration target. Fitting the gain directly is well calibrated: binned over the held-out fold, predicted and realised gain agree to 0.029 (Fig. 10), against a correctness ECE of 0.027 for the same signal, so the correction costs nothing in calibration quality.

Does removing the knob cost anything? It should not, if the rule is right. Sweeping V over 100 to 3200 ms per correct answer places six operating points on the accuracy-latency plane without any rate ever being chosen; five of the six are undominated by the tuned sweep of Fig. 6 evaluated on the same fold (Fig. 11), and 78% remain undominated across the 200 resplits. An untuned rule reaches a tuned frontier, which is the useful form of this result: the rate was never information, only a stand-in for a preference the operator already has.

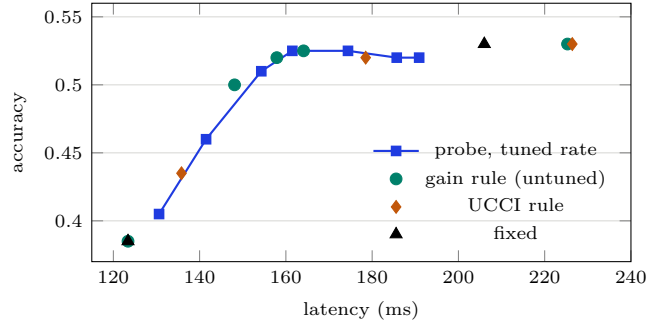


Figure 11: Untuned decision rules against the tuned sweep. The gain rule tracks the swept frontier without a rate parameter; the UCCI rule, calibrating correctness instead, sits inside it and collapses to always-escalate once V is large.

Where the rule refuses. At $V \leq 200$ the break-even of Eq. 11 exceeds any attainable gain and the rule escalates nothing, returning the always-cheap policy. That is the correct answer, not a degenerate one: by Eq. 10 no expected gain on this pilot repays 102.9 ms when a correct answer is worth 200. A tuned rate cannot express this, which is the failure mode reported for our earlier threshold tuners, they discovered “never escalate” on one fold and had no way to say whether that was a preference or an artefact. Where a floor on escalation is required instead, [12]’s recall-controlled gates give one that cannot degenerate by construction.

7.1 The rule transfers where the signal does not

Every number above uses the probe, which §6 discredits within a domain. So we re-ran the whole comparison on output entropy, the signal that does work within a domain, fitted and tested inside each dataset over 40 resamples, with escalation priced per example. If the rule is a property of the method rather than of the signal, it should survive.

It does, and the confound makes it more useful rather than less. Escalation value is domain-determined, so a fixed rate is wrong in every domain but the one it was tuned on. A rule calibrated on the observed gain distribution picks its own rate wherever it is fitted, from one unchanged operator preference (Fig. 12).

Fixing one policy and applying it to all four domains, weighted by dataset size, the rule reaches 0.489 accuracy at 181.6 ms where a tuned 30% rate reaches 0.464 at 195.1 ms: more accurate *and* cheaper. Tuned rates at 40% and 50% are dominated as well. Only the two lowest tuned rates survive on the front, at the cautious end where escalating little is right everywhere.

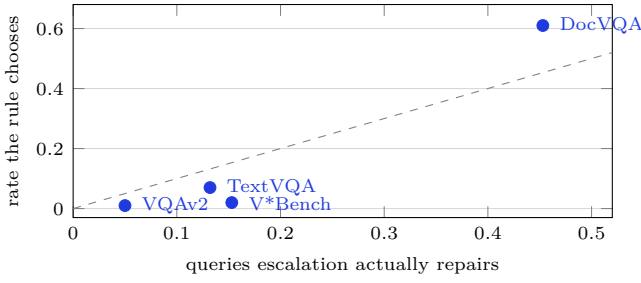


Figure 12: One value of V , applied unchanged to four domains. The rule escalates 61% of DocVQA and 1% of VQAv2, tracking how much escalation actually repairs there ($r = 0.974$). The dashed line is not a target, escalating every repairable query is not optimal, but it fixes the scale.

7.2 How far to escalate, not just whether

Every escalation method we have read is binary: answer from a downsampled image, or run the full one. Taking the cheapest configuration that answers each query correctly says that collapses the decision carrying most of the money.

cheapest rung that answers correctly	share
384 px (no escalation)	38.3%
intermediate rung	17.7%
full resolution	5.2%
nothing we run	38.8%

Of the queries escalation can repair, 77% are repaired by the intermediate rung. A binary policy over-serves three quarters of what it escalates. The rungs are also not equally efficient: $384 \rightarrow 768$ buys 13.3 points for 79 ms (+1.69 points per second) where $768 \rightarrow$ full buys 3.3 for 38 ms (+0.86). The binary jump measures +1.42, which is what averaging a good rung with a poor one gives.

The rule generalises without modification. Rung r has its own expected gain over answering cheap and its own extra latency, so

$$r^*(x) = \arg \max_r (V \mathbb{E}[G_r | x] - \Delta t_r), \quad (12)$$

answering cheap when no rung is positive. Eq. 12 reduces to Eq. 11 when there is one rung; the binary policy is the special case that has deleted the middle of its own ladder. One calibrator per rung is fitted on the same score, so the additional cost is a second isotonic fit over cached values.

Proposition 1 extends without change. Escalating to rung r shifts expected cost by $\Delta J_r(x) = -w_e \mathbb{E}[G_r | x] + \lambda_t \Delta t_r$, the choice is again pointwise because the objective is linear in the selection, and the best action at x is the rung minimising ΔJ_r , or the cheap pass when every ΔJ_r is non-negative. Multiplying through by $-1/\lambda_t$ gives Eq. 12. The binary rule is therefore not a simplification of the ladder but a *constrained* version of it, forced to choose between two of the rungs available to it.

Algorithm 2 Calibrated ladder escalation

```

fit, given rungs  $r = 1 \dots R$  in increasing cost
1  $s_i \leftarrow$  any routing score read from the cheap pass
2 for each rung  $r$ :
3    $G_i^{(r)} \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^r] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^r]$ 
4    $\hat{g}_r \leftarrow 2 \text{isotonic}(s, (G^{(r)} + 1)/2) - 1$ 
serve  $x$ , given  $V$ 
5  $\Delta t_r \leftarrow t_r(x) - t_{\text{cheap}}(x)$  for each rung
6  $v_r \leftarrow V \hat{g}_r(s(x)) - \Delta t_r$ 
7 if  $\max_r v_r > 0$  then run rung  $\arg \max_r v_r$ 
8 else answer from the cheap pass

```

	top rung exists	accuracy	latency
DocVQA	98%	+0.003	−35.1
V*Bench	98%	−0.008	−3.5
TextVQA	0%	−0.006	+1.2
VQAv2	0%	−0.002	−0.8

Table 15: Ladder minus binary, averaged over four operator preferences, latency in milliseconds. “Top rung exists” is the share of images on which the full-resolution pass spends more visual tokens than the intermediate one.

It pays where the ladder physically exists, and nowhere else. Paired over four domains and four preferences, the ladder is -9.6 [$-17.2, -2.6$] ms cheaper at -0.003 [$-0.008, +0.002$] accuracy: significantly cheaper, indistinguishable in accuracy. The per-domain breakdown is the real result (Table 15).

On DocVQA the ladder saves 35 ms at no accuracy cost. On TextVQA and VQAv2 it does nothing, correctly, because for those datasets there is no top rung to find.

A configuration name does not determine a configuration. That last column is a measurement fact worth stating on its own. The processor caps its target at the input’s longest side, so on a small image the “full-resolution” pass *is* the intermediate pass: identical token count, identical cost, identical output. This holds for 100% of TextVQA and VQAv2 and 2% of the other two, 56% of the pilot. The escalation this paper studies is a real escalation for half its data.

This is the fourth error of the same family we have found, after processor upscaling, patch-grid quantisation, and the flat escalation price of §6. All four come from treating a configuration label as a cost or a capability. The general form of the lesson: in adaptive-resolution work, verify that the configurations differ *on the data*, not in the config file.

A ladder built on measured rungs, and where it stops. The mixture cannot price the second and third rungs, because for over half of it they are the same configuration. So we collected a single-domain pilot designed against that: 1200 DocVQA pages, with rungs chosen by

step	net gain [95% CI]	cost	points/s
64 → 320	+0.383 [0.352, 0.414]	105	+3.65
320 → 640	+0.086 [0.060, 0.111]	129	+0.67
640 → 1088	−0.002 [−0.026, 0.022]	156	−0.01

Table 16: Rung-to-rung value on 1200 DocVQA pages, by visual-token count, latency in milliseconds. Accuracy peaks at 640 tokens (0.748) and 1088 tokens reaches 0.746: overlapping intervals for 70% more tokens.

measuring SmolVLM’s patch-grid buckets (64, 320, 640 and 1088 visual tokens) rather than by picking round pixel sizes. Every rung is strictly dearer than the one below it on 95 to 100% of images, and latency is affine in tokens at $205.0 + 0.415v$ ms with a worst residual of 20.2 ms.

What the taxonomy looks like on paper. Figure 13 grounds the shares the ladder trades on, one real page per outcome. The thumbnail suffices on 27.9% of the 1200 pages, a higher rung flips a wrong answer to a right one on 49.9%, and 13.0% fail everywhere, full resolution included: the third panel fails by reading the wrong table row, an error more pixels cannot repair. The remaining 9.2% are non-monotone in the rung order, correct at 768 px or at full resolution while 1152 px fails.

The result is a saturation point, and it arrives early (Table 16).

The first rung is *five times* more efficient per millisecond than the second, and the second is the last one that pays: the top step repairs 8.1% of queries and damages 8.3%, a wash costing 156 ms. Of everything escalation can repair, 93% is repaired below the top rung and only 4.2% needs it. And this is measured on document pages, the regime where resolution matters most, so it is not a low-detail artefact.

More pixels stop buying accuracy well below the resolution the model accepts. That is a sharper form of the non-monotonicity in §3.3: escalation does not merely harm a few queries, it has a ceiling, and past it the harm rate catches the repair rate exactly.

A null result is only a result if it is tight, so it is worth saying that this one is. The interval on the top step has half-width 0.024, against measured gains of +0.383 and +0.086 on the steps below it: an effect a third the size of the smallest one we do detect would have been visible. That distinguishes this from a claim of no effect made at a sample size incapable of finding one, which is the failure mode we impose on ourselves elsewhere and check for here.

The ceiling belongs to the corpus, not to the model.

A saturation point measured on one model could be that model running out of capacity, or the pages simply being legible at that resolution. The two explanations predict opposite things, so we re-ran the entire ladder on

	384	768	1152	2048	top step
SmolVLM-500M	0.279	0.662	0.748	0.746	−0.002
SmolVLM-256M	0.220	0.588	0.653	0.666	+0.013
SmolVLM2-2.2B	0.413	0.669	0.738	0.754	+0.016
Qwen2-VL-2B	0.600	0.850	0.890	0.892	+0.002

<i>median visual tokens spent at each rung</i>					
SmolVLM-500M/256M	64	320	640	1088	
SmolVLM2-2.2B	81	405	810	1377	
Qwen2-VL-2B	154	567	1312	4161	

Table 17: ANLS accuracy by rung, four models on the same DocVQA pages (1200 for the SmolVLM family, the first 500 for Qwen2-VL-2B), columns labelled by the pixel target. Every top-step interval is a tight null: $[-0.026, +0.022]$, $[-0.012, +0.037]$, $[-0.011, +0.041]$ and $[-0.016, +0.020]$, half-widths 0.018 to 0.026.

SmolVLM-256M over the same 1200 pages: capacity implies the smaller model saturates *earlier*, legibility implies the same rung with lower accuracy throughout (Table 17). A reviewer of an earlier draft made the obvious objection to the three models we had then, and it was right: they share a lineage and a data recipe, so a common pretraining corpus was an untested alternative cause. The fourth row answers it.

It is legibility. Halving the parameters costs accuracy everywhere and moves the ceiling nowhere, and all four top-step intervals are tight, so these are measured nulls rather than failures to measure. Plotted, the four curves rise and then flatten at the same place (Fig. 14).

The ceiling survives leaving the family. Qwen2-VL-2B shares nothing with the SmolVLM line: a NaViT-style encoder that emits tokens in proportion to input pixels rather than quantising to a patch grid, a different language model, a different data recipe. It is also a far stronger reader of these pages, peaking at 0.892 against the family’s best 0.754. That matters for what the null can mean. A model that stops gaining because it is too weak to exploit more pixels would prove nothing; this one stops gaining while answering 14 points more of the corpus correctly than any model in the family, and while being handed $3.2\times$ the visual tokens to do it with (1312 at the ceiling, 4161 at full resolution) for a measured $+0.002 [-0.016, +0.020]$. Whatever is left unanswered at 1152 px on these pages is not waiting on resolution for any of the four models, which points at the documents rather than at the models reading them.

The ceiling is not a token budget. The two SmolVLM models share an 86M vision encoder, so “stops gaining at 1152 px” and “stops gaining at 640 visual tokens” name the same point for both and cannot be told apart. SmolVLM2-2.2B tokenises differently, spending 810 tokens where the others spend 640, which separates them, and Qwen2-VL-2B spends 1312 there, a $2.05\times$ spread across the four. All four stop at the same *pixel* target

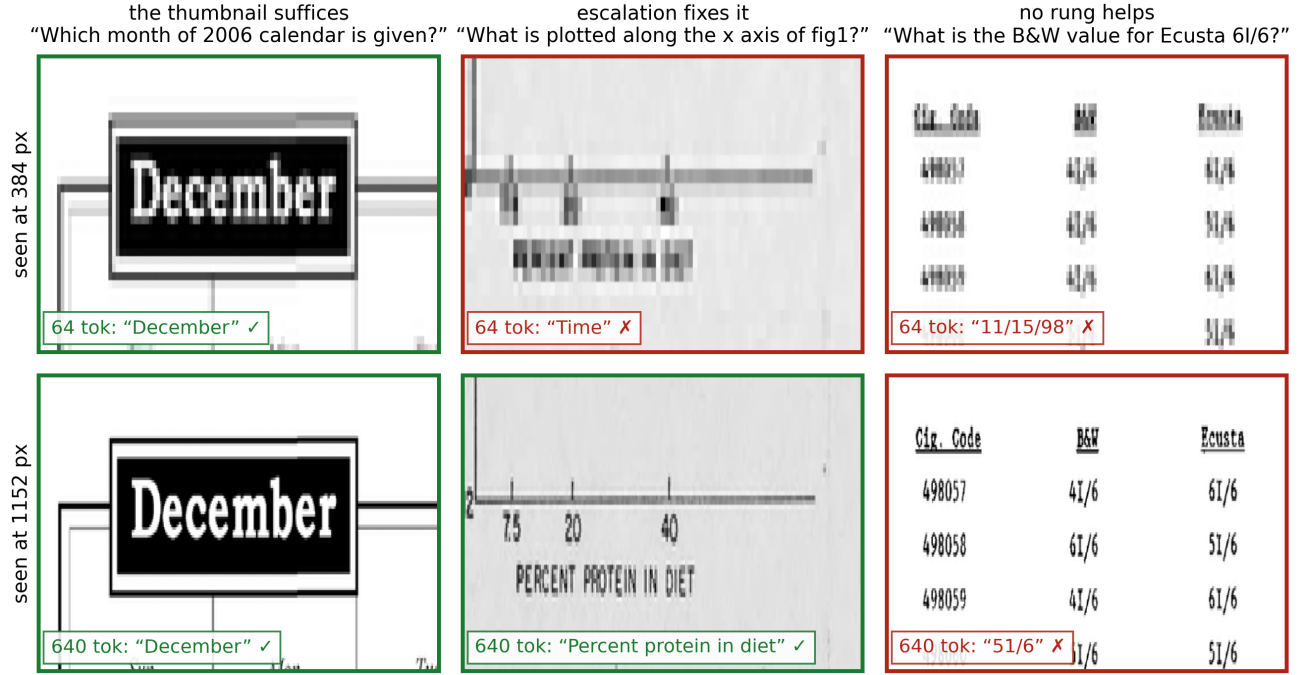


Figure 13: Three DocVQA pages rendered at the resolution each rung actually receives (Lanczos resize to the rung’s longest edge, shown nearest-neighbour; the model’s answer and its visual-token price are inset). Left: layout-scale evidence survives the thumbnail and 64 tokens already answer; this covers 27.9% of the 1200 pages. Centre: the axis label is an unreadable smudge at 384px and the model confabulates “Time”; at 1152px it reads. A higher rung repairs a wrong answer this way on 49.9% of pages. Right: the table resolves at 1152px yet the model reports the value of the wrong row, and full resolution repeats the error; 13.0% of pages fail at every rung, and for them the binding constraint is not legibility, so no resolution policy can help.

while spending very different numbers of tokens to reach it, which no single sequence-length ceiling explains.

But it is not purely a pixel count either, and one model says so. The obvious next sentence, that what runs out is the information in the pixels, is the one we wrote in an earlier draft and cannot support. In all four models the two quantities travel together: raising the pixel target raises the token count, so every rung changes both, and the argument above rules out a *fixed* token budget without establishing that tokens do no work. Separating them needs a model that spends the same tokens on a small image as on a large one, and InternVL3-1B is one by accident of its design. It selects a patch grid from the aspect ratio rather than the resolution, so a page downsampled to 384px is upscaled again to fill the same tiles as the 2048px original. Across 500 pages all four rungs are token-identical on 95% of them and the mean spend moves by $1.02\times$ against $5.3\times$ in input pixels: near enough to constant to read the ladder as pixels alone (Table 18).

With the budget held still, the gain stops one rung earlier, at 768px. The $768 \rightarrow 1152$ step, worth $+0.086$ to SmolVLM-500M and $+0.040$ to Qwen2-VL-2B, is worth -0.008 $[-0.026, +0.010]$ here. Two readings are open and

step	net gain [95% CI]	tokens	verdict
384 $\rightarrow$ 768	$+0.108$ $[+0.070, +0.148]$	+16	gain
768 $\rightarrow$ 1152	-0.008 $[-0.026, +0.010]$	+20	null
1152 $\rightarrow$ full	$+0.014$ $[-0.002, +0.030]$	+38	null

Table 18: InternVL3-1B on 500 DocVQA pages, where the token budget is near constant and only the input pixels move. Accuracy is 0.692, 0.800, 0.792, 0.806 by rung. Both nulls are tight, half-widths 0.018 and 0.016.

one model cannot close them: either the pixel information of these pages is exhausted by 768px and what the other four bought in that step was their growing token budget, or InternVL simply reads less of the page. The second is the weaker of the two, because InternVL is not the weaker model. It peaks at 0.806 against SmolVLM-500M’s 0.748, so a model that reads *more* of this corpus overall is the one that stops gaining from pixels first.

The other axis, with the pixels held still. The control above moves pixels and holds tokens; the pair is only complete with its converse. InternVL selects its tile grid under a bound, so raising the bound multiplies the visual tokens spent on an image whose pixels never

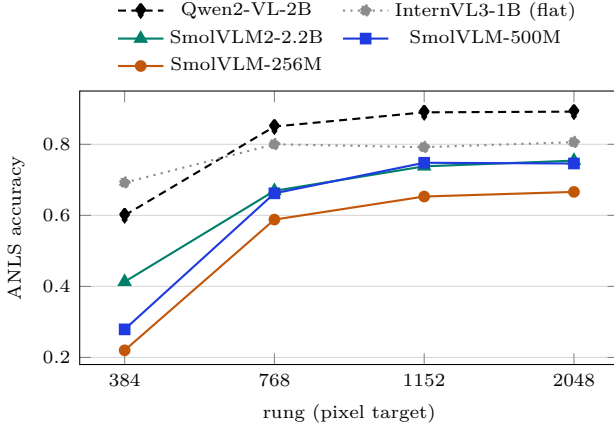


Figure 14: Accuracy against rung on the same DocVQA pages. The last step is flat for the four cost ladders, though they spend 640, 640, 810 and 1312 visual tokens respectively at the rung where it flattens, so no fixed token budget explains the ceiling, and the dashed curve is a model from another lineage entirely. The dotted curve is the control: InternVL3-1B spends the same tokens at every rung, and flattens one rung earlier, at 768 px. Its vertical position is not comparable to the others, since its cheapest rung already spends a full budget on an upscaled image.

change. Pinning the input at full resolution and sweeping the bound over 500 pages measures what sequence length alone buys here (Table 19).

Tripling the token budget at fixed pixels buys $+0.024 [-0.006, +0.054]$, a null tight enough to have found an effect a third the size of the pixel gains we do detect. Spending a further 2258 tokens is associated with 0.046 *less* accuracy where it is actually spent; that sign is nominally significant and does not survive the family-wise correction of §9 ($p = 0.023$, adjusted 0.372), so we report it as an observation consistent with the premise of the token-pruning literature [4] and with a mechanism of interpolation rather than information, and not as an established effect. The pages where the bound changed nothing form an internal control that behaves exactly as greedy decoding requires, $+0.000 [+0.000, +0.000]$ over 260 pages, which bounds the measurement noise in the other subset.

Read against the fixed-budget ladder, this separates the two quantities. Holding tokens still and raising pixels from 384 to 768 buys $+0.108$, which survives correction; holding pixels still and tripling tokens buys $+0.024$ with an interval spanning zero, and spending beyond that does not help. On this corpus the binding constraint is the information in the pixels, and sequence length is not scarce. That is the sentence an earlier draft asserted without evidence, then withdrew for want of it, and it is now supported by a pair of controls rather than by the co-varying ladders that cannot speak to it.

step	pages	extra tokens	net gain [95% CI]
1280 → 3328 tokens	500	+2011	+0.024 [-0.006, +0.054]
<i>next bound, split by what it actually spent</i>			
tokens rose	240	+2258	-0.046 [-0.087, -0.008]
tokens unchanged	260	0	+0.000 [+0.000, +0.000]

Table 19: InternVL3-1B on 500 DocVQA pages with the input pinned at full resolution and only the tile bound moving. Accuracy is 0.782, 0.806, 0.784 by bound. The second bound raises the token count on 48% of pages and leaves the rest untouched, so the step is reported conditioned on what was spent rather than on the bound’s name.

What we are entitled to say is therefore narrower than what we first wrote, and better supported. No single token budget explains the ceiling; pixel information alone is spent by 768 px on the model that lets us isolate it; and additional sequence length, isolated in turn, does not repay its cost on this corpus. What we cannot do is attribute the 768 to 1152 px gains of the four co-varying models between the two axes, since both move at every rung there. The serving consequence is untouched either way, because it never depended on the mechanism: every model we tested stops paying by 1152 px, and Algorithm 3 finds that point without needing to know why it is there.

That makes the saturation point reusable, as a bound. Across these four it survives a change of serving model, of vision encoder and of lineage, which is already a more useful object than a threshold re-derived per model. It is not a constant, and the fifth model is why: with the token budget held flat the ceiling sits a rung lower, at 768 px. The direction is the part worth keeping. No model we tested needs *more* resolution than the corpus ceiling, and one needs less, so the ceiling measured this way is an upper bound on what any model needs and running Algorithm 3 against the model actually being served can only move it down. A serving stack that caps escalation at the corpus ceiling is never wrong about the top of its ladder, only sometimes generous. We note in fairness that three of the four put a positive point estimate on the top step ($+0.013$, $+0.016$ and $+0.002$), leaning the opposite way from the 500M’s -0.002 ; at these interval widths that is noise, and we report it rather than claim a symmetry the data does not show.

The procedure, and what it costs to run. If the ceiling is the transferable object then it has to be findable on a corpus nobody has measured, so we state the procedure we followed rather than only its output (Algorithm 3). Two of its steps encode failures this paper documents elsewhere. Step 2 measures the visual tokens each pixel target actually produces instead of trusting the target, because four measurement errors here arose from a configuration name that did not determine a cost, and on our own pilot two nominally different rungs were the same configuration

Algorithm 3 Locating a corpus’s resolution ceiling

- 1 draw n pages from the corpus ($n \geq 300$, Table 20)
- 2 for each candidate pixel target p , *measure* the visual tokens the serving model spends on each page at p
- 3 keep the targets whose token counts are strictly increasing on $\geq 95\%$ of pages; discard the rest as duplicates
- 4 run every kept rung; record correctness and latency
- 5 for each adjacent pair (r, r') , bootstrap the paired gain $\bar{G}_{r \rightarrow r'}$ and its 95% interval
- 6 **if** any interval has half-width > 0.05 **then**
report undetermined and collect more pages
- 7 ceiling $\leftarrow$ highest r' whose interval excludes 0
- 8 serve with every rung above the ceiling removed
(run on the served model: the corpus ceiling bounds it above)

pages	half-width	p90	names the same ceiling
100	0.080	0.090	47%
200	0.055	0.062	80%
300	0.045	0.050	93%
500	0.035	0.038	100%
800	0.028	0.029	100%
1200	0.023	0.024	100%

Table 20: What a sample size buys when locating the ceiling, 200 draws per row on the 1200-page pilot. Half-width is of the paired top-step interval; the last column is how often a run at that size names the rung the full corpus names.

for 56% of examples. Step 6 refuses to declare a ceiling from a wide interval, the failure mode that produced a premature verdict at $n = 244$ on an earlier pass and again at $n \approx 100$ mid-collection on the out-of-lineage run.

The cost of running it is small and measurable. Sub-sampling the pilot and recomputing the top-step interval gives the precision a given sample buys (Table 20): the half-width falls as $0.78/\sqrt{n}$, so the 0.05 bar of step 6 is reached at about 244 pages, and 300 pages name the same ceiling rung as the full 1200 on 93% of draws. The row that matters most is the first. At $n = 100$ the procedure agrees with the full corpus barely half the time, which is not a ceiling measurement but a coin flip, and it is exactly the trap a practitioner would walk into by eyeballing a small pilot.

The procedure on a corpus it has not seen, and a prediction it refutes. An earlier version of this paper predicted that a benchmark of denser type should place the ceiling higher than DocVQA’s 1152px. We collected one and ran Algorithm 3 on it unchanged: 500 InfographicVQA pages, the same task, the same ANLS metric, the same pixel targets, and pages carrying a far wider range of type sizes (Table 21).

The prediction fails and the claim it was attached to survives in a stronger form. The ceiling does not move: both corpora stop paying at 1152px, and on both

	384	768	1152	2048	ceiling
DocVQA ($n=1200$)	0.279	0.662	0.748	0.746	1152
InfographicVQA ($n=500$)	0.230	0.282	0.368	0.360	1152
<i>median visual tokens spent at each target</i>					
DocVQA	64	320	640	1088	
InfographicVQA	64	192	320	576	

Table 21: The same procedure on two corpora. The top steps are -0.002 $[-0.024, +0.022]$ and -0.008 $[-0.044, +0.028]$, both inside the precision bar of Algorithm 3’s step 6. The $768 \rightarrow 1152$ step pays on both, $+0.086$ each.

the top step is a measured null rather than an absence of measurement. What moves is the price. The same pixel target costs 640 visual tokens on DocVQA and 320 on InfographicVQA, whose pages are tall and narrow and fill fewer patches at a given longest edge, so two corpora reach an identical ceiling while spending a factor of exactly two apart. Read alongside the cross-model result above, the ceiling is invariant to the serving model at fixed corpus and to the corpus at fixed pixel target, while the token count at that ceiling varies by $2.05\times$ across models and $2.00\times$ across corpora. **A token budget cannot be what runs out if two corpora exhaust it at half each other’s spend.**

We report the failed prediction rather than deleting it. It was the reason to collect this corpus, and its failure upgrades “1152px is DocVQA’s number” to “1152px is where two document corpora stop paying”. That is wider than we were entitled to before and narrower than universality: both are document-QA, and a corpus of genuinely finer print remains untested.

One thing does not transfer with it. On InfographicVQA the model is far weaker (0.368 against 0.748 at the ceiling), and the ladder’s advantage over the hull nearly disappears: it clears at one operator preference of four ($+0.006$ $[+0.002, +0.011]$ at $V=800$) rather than at three. The free image descriptor fails on this workload as on the other, at every preference, which is a second confirmation of the inversion in §5.1. Where a model is weak enough that most queries are unanswerable at any rung, there is little for any per-query policy to allocate.

The ladder rule finds the ceiling on its own. Given rungs and prices, Eq. 12 selects the top rung on 0 to 6% of queries across the whole preference range: it discovers the saturation from the calibrated gains without being told. Paired against the binary policy over 30 resamples of a held-out 300 (Table 22), it is significantly cheaper at three of four operating points, and at the highest preference it is cheaper at accuracy the comparison cannot separate.

The mixture measured this effect at -9.6 ms because most of its ladder did not exist. We report the intervals per operating point rather than pooled across them: an earlier version of this comparison bootstrapped the

V	accuracy delta	latency delta (ms)
400	+0.359 [+0.346, +0.374]	+86.0 [+84.3, +87.4]
800	-0.010 [-0.019, -0.001]	-161.1 [-163.9, -158.4]
1600	-0.010 [-0.017, -0.003]	-154.5 [-158.3, -150.7]
3200	-0.002 [-0.009, +0.005]	-136.4 [-141.1, -132.0]

Table 22: Ladder minus binary on 1200 DocVQA pages, paired within each operating point over 30 resamples. At $V = 400$ the binary rule barely escalates at all and the ladder buys 36 accuracy points for 86ms; above that it trades at most one accuracy point for roughly 150 ms.

four preference means, which resamples four numbers and yields an interval four points wide.

A refinement of ours that does not pay. The cost audit suggested an obvious improvement: since escalation prices vary $2.5\times$ across the pilot, charge each query its own break-even $\tau_i = \Delta t_i / V$ rather than a global one. Implemented and measured, it is worth nothing, +0.00002 [-0.0003, +0.0004] accuracy and -1.5 [-4.3, +0.4] ms, both spanning zero.

The reason is the confound again. Image size is largely a *domain* property, so within a domain the prices a per-query rule would discriminate between barely differ, and there is nothing for the refinement to bite on. It would matter for a router serving one queue of genuinely mixed image sizes, and we report it as a negative rather than dropping it because the negative is the informative part: the same fact that makes the probe a domain detector makes per-query pricing redundant within a domain.

One caveat on the UCCI contrast. The gap of Table 14, calibrating correctness over-spending against calibrating the gain, is a mixture effect too. Within a domain both rules land on the front at different operating points, and the sign flip that separates them is weaker. We state the contrast as measured on the mixture and unresolved within a domain.

8 Discussion

What an evaluation has to include before a signal can be believed. The measurements above converge on three requirements, each of which changed a conclusion here when applied. Score signals *within* each distribution rather than across a mixture: pooling inflated our probe from a within-domain 0.572 to a pooled 0.760, and the inflation is the mixture’s own heterogeneity. Charge every signal what it costs to read: entropy ranks respectably and still fails the no-signal baseline, because reading it requires finishing the pass it is meant to price. And compare against randomisation between fixed configurations, not against a single fixed endpoint, since the endpoints are the two least interesting points of a reachable segment. None of

the three is expensive. Together they are the difference between the result we first believed and the one we can defend.

What this means for the systems we compare against. It bounds rather than refutes them. The trained policies of [1, 2, 3] operate at 7B, where the untrained self-report we can measure is worth little (Table 6) and where the training is presumably what closes that gap. Our finding is about what is available *without* training below one billion parameters, and there the honest answer on mixed traffic is that the file header is competitive with the residual stream. That is a statement about the difficulty of the sub-1B regime, not about the value of learning the decision. It does suggest one cheap control for that literature: report the escalation policy’s accuracy against a policy that knows only the benchmark identity, which on our mixture accounts for the whole of the adaptive gain.

Where the remaining money is. Not in deciding *whether* to escalate on heterogeneous traffic, where a free descriptor suffices, and not in a third abstention action, which the calibrated rule already suppresses (§E). It is in deciding *how far*, which is worth 35 ms per query at no accuracy cost on document traffic, and in knowing where to stop, which is a property of the corpus rather than of the query and needs measuring once. A per-query signal earns its keep exactly where domain identity carries no information, and there it is output entropy rather than the probe that survives. The practical rule is therefore a two-line one: measure how much of the escalation value your traffic explains with free metadata; if most of it, route on the metadata and spend the engineering on the ladder; if little, read the model and give it graded actions.

Status of each claim. Table 23 separates what the evidence supports at three different strengths, since the paper’s claims are not all of one kind.

9 Limitations

Five serving models, one GPU, and two pilots: 1000 examples across four datasets and 1200 within one, the latter run on all five. The saturation ceiling is the claim we have pushed hardest, over an $8.6\times$ parameter range, three vision-encoder mechanisms and two lineages. It held across four of the five and moved down a rung on the fifth, the one whose token budget does not follow its input resolution, which is why we state it as a bound rather than a constant. It was measured on two corpora, DocVQA and InfographicVQA, which stop paying at the same pixel target while spending a factor of two apart in visual tokens to reach it. Both are document-QA, so the claim we are entitled to is that document pages of this kind have a legibility ceiling near 1152px that

bounds every model’s useful resolution from above and that Algorithm 3 locates in about 300 pages. We predicted that the denser of the two would place the ceiling higher and it did not, which is reported in §7.2 rather than removed. A corpus of genuinely finer print, or of a different medium, remains untested, and the procedure rather than the number is what we claim transfers. The mechanism behind the ceiling is settled only on the model that lets both axes be moved separately. There, pixels at fixed tokens buy +0.108 over one rung and tokens at fixed pixels buy +0.024 with an interval spanning zero. We do not claim the same decomposition for the four models whose tokeniser follows resolution, because every rung of theirs moves both quantities, and a corpus with finer print could shift the balance. Our strongest baseline is the published escalation *mechanism* without its training; a re-implemented VisionThink at sub-1B scale remains the comparison the field would want, and the gap between an untrained self-report and a trained one is exactly what we cannot measure.

Four limitations deserve more than a mention. The latency saving needs an inference stack that can abandon a prefill mid-flight and start a different image. We verified this is real rather than arithmetic: physically truncating the decoder to its first six layers gives a 20.5 ms prefill, within 1% of Eq. 7’s prediction, and an escalated query then costs 277.0 ms against 402.9 ms for completing the cheap pass first, a measured 31% saving. What standard serving frameworks do not expose is the *control* to do this per query, so the saving is available but not yet convenient. Absolute accuracies run from 0.385 to 0.580, a regime where the model is wrong about half the time, and conclusions drawn there may not transfer to a model that is usually right. The probe’s direction does not transfer across question domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) so deployment needs a domain-representative calibration set and a shift detector. And the cross-model transfer result uses two models that share an identical 86M vision encoder, so part of what transfers may be encoder-side rather than genuinely cross-model; separating those requires a source model with a different encoder.

We ask many questions of one dataset mixture, so the intervals above are corrected rather than nominal. Sixty-three paired comparisons at the nominal 5% level carry a 96% chance of at least one false positive, so we convert each bootstrap to a two-sided p-value and run Holm’s step-down over the whole family. Every comparison carries its own resample vector, so all sixty-three are scored by the same bootstrap rather than by an approximation to a reported interval.

Forty-seven of the sixty-three clear the nominal level and thirty-nine survive the step-down. Eight claims are lost, and they are worth naming individually because four of them are convenient for our thesis. The probe’s AUROC advantage on the recovery target adjusts from $p = 0.041$

to 0.662. Two preferences at which the probe nominally beats the free image descriptor adjust to 0.612 and 0.426, which makes the two signals harder to distinguish rather than easier. Two of the points at which the free descriptor sits *below* the hull inside one workload adjust to 0.130 and 0.372, which weakens that row to indistinguishable from the hull rather than worse than it: the correct reading is that the descriptor buys nothing there, not that it is harmful. Both readings of the harmful top step in the tile-budget control adjust to 0.745 and 0.722. And the second corpus’s weakest rung gain, +0.052 from 384 to 768 px on InfographicVQA, adjusts to 0.331; the step that sets that corpus’s ceiling, +0.086 from 768 to 1152 px, survives, so the ceiling determination does not depend on the lost one. Every cost comparison survives, as do the pixel-axis gains, the domain-identity bound, the free descriptor’s clearance of the hull at every operating point of both costings, and the entropy ladder’s clearance inside the DocVQA workload. We keep the eight in the paper as observations and no longer state them as established.

The split is itself informative. **Every surviving comparison is about cost**; all three probe-versus-entropy *accuracy* comparisons fail, as does the top rung’s gain. The split matches the paper’s claim structure: what we demonstrate is cheaper compute at accuracy the data cannot separate, not better answers.

The Pareto-dominance result is the one we have tested hardest and it came back qualified: universal dominance holds on the reported fold and in 52% of resplits, so the claim we make is the distributional one (90% of the front, a saving positive in 200 of 200 splits) and not the categorical one. Readers should treat any single-fold dominance statement in this literature, including our own earlier one, as untested until resampled.

One limit is specific to §7: V is elicited, not measured, so the contribution is that the policy is determined *given* a preference, not that we know the preference. The adversarial result is a proxy: it shows the probe is not incidentally moved by what moves entropy, not that it resists an attack optimised against it.

The confound audit of §6 has a limit of its own, in our favour and worth stating. Fitting inside one dataset gives the probe 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. Two observations argue against that reading (entropy needs no training at all and is therefore unaffected, and the probe is at or below chance rather than merely noisy) but the clean test is a single-domain corpus large enough to fit and test inside, and 300 examples per dataset is not it. We report the confound as established and its magnitude as uncertain.

10 Reproducibility

Every numeric claim in this paper is re-derived from the run records by a validation harness with an explicit thresh-

old per claim, which exits non-zero if any stops holding; it currently checks sixty-eight, including the negative results on localization, on our own energy instrument, and on the universality of Pareto dominance, that last one asserts the claim is *not* universal, so restating it too strongly fails the build. Claims stated in the text and not covered by a check are marked as unverified. The oracle runs are stored as append-only JSONL with per-configuration answers, generation scores and hardware measurements, so correctness metrics and cost weights can be changed offline without re-running any model. Splits are hash-based on example identifiers, so growing the pilot never reshuffles examples already assigned. Decoding is greedy throughout (16 new tokens, no sampling), so every accuracy in this paper is deterministic given the checkpoint and the image. The single-domain pilots time each pass once; the cost model is fitted on per-bucket medians over at least 50 passes, so no individual timing can set the line. The discovery that 56% of the mixture’s full-resolution passes were not distinct configurations (§7.2) is a protocol defect of ours that the verified ladder of the single-domain pilot was designed to remove.

Compute. Every experiment in this paper runs on one RTX 4060 laptop GPU with 8.6 GB of memory. The oracle runs comprise 54660 instrumented generations totalling 305 minutes of measured model time, of which the 1200-page single-domain pilot is 6000 generations per serving model, 40 minutes each for the two SmolVLM models and 54 for SmolVLM2-2.2B; activation extraction and the component profiles add roughly a further half hour. Wall-clock is appreciably larger, since image decoding, the processor, and the instrumentation itself sit outside the timed region. No training of any kind was performed: every probe here is a difference of means over cached activations, and every calibration a single isotonic pass, both of which take seconds on a CPU. The reproduction cost of this paper is therefore about six GPU-hours, which is the point of working at this scale.

11 Conclusion

We set out to ask what a 500M vision-language model knows about its own need for more pixels before it generates anything. Measured against baselines that need no model at all, the defensible answer is narrow, and the baselines are the reason it is worth stating.

On a four-dataset mixture, a linear probe on pre-generation activations ranks escalation value better than output confidence and yields a policy readable after a sixth of the language model. Inside any one of those datasets it reaches 0.572 where output confidence, fitted on nothing, reaches 0.663, and more training data narrows that gap without closing it. What the direction mostly encodes is which *kind* of image is present: a quantity that predicts escalation value across a benchmark mixture, because the

mixture’s datasets differ enormously in how much resolution buys, and predicts much less within one. Raw image size, free to read, carries the same information: built into a policy and priced, it clears the randomisation hull at every operating point of either costing and stays within 0.010 of the probe at every operator preference. A policy handed the dataset label and escalating at random within each dataset dominates both, which places an upper bound on the whole exercise: on mixed traffic, the adaptive gain available is the gain from knowing which benchmark is being served.

Three things bound the claim rather than one. This is a way of deciding *whether* to look again and not *where*; on the evidence here it is a way of deciding for a heterogeneous mixture and not for a workload; and on a mixture it is not a way that needs the model’s activations at all. Inside a workload the same measurement inverts: the free descriptor falls to chance and only a model-read signal on a graded action space clears the no-signal baseline. Neither regime is the general case, which is the result we would put first.

What we would keep is the part that did not depend on the probe. The decision rule is signal-agnostic and survives on output confidence, where one stated preference produces a different escalation rate per domain without retuning. The resolution ladder carries most of the money that was attributed to the escalate/answer decision: on document pages, 93% of what escalation repairs is repaired below the top rung, and the ceiling sits at a pixel resolution that four models with three tokenisers and two lineages agree on. A fifth, whose token budget does not follow its input resolution, stops one rung earlier, so that ceiling is the most any model needs rather than what each one needs.

The comparison that decides whether any of this matters is against randomising between fixed configurations, which needs no signal. The standard entropy threshold never clears that bar on our mixture; the probe clears it only because it is read mid-prefill; the ladder clears it on homogeneous traffic. Where a single fixed rung fits the budget, pinning it at the corpus’s measured saturation point is the right policy, and per-query routing is for the traffic no single rung fits. Both need the ceiling measured rather than guessed, which is why we state the procedure and its price alongside the number.

The result we would most like others to carry away is the negative one in §5.1, which we encountered four times. A probe trained on the conditional target beat its baseline on ranking and lost the policy comparison by 8.5 points. An ensemble of entropy and probe then beat both on ranking and lost the policy comparison on cost. A rule calibrating the weak pass’s error probability, which is the standard cascade recipe, matched our accuracy while spending 59% relatively more compute. And the signal that beat all of them on the mixture turned out to be ranking the mixture. The target was wrong, then the signal was too expensive to read, then the calibration

target was wrong, then the evaluation distribution was doing the work. None of the four is visible in an AUROC table.

Reporting ranking quality without building the policy is reporting the wrong number. The four ways of getting it wrong are optimising a target the policy does not face, optimising a signal the policy cannot afford, calibrating a quantity whose assumptions the setting violates, and reading a between-group difference as a within-group one. All are easy to miss.

The last is the one we would flag most broadly, with the caveat that the literature we borrowed the probe from had already avoided it: any probe evaluated on a mixture of benchmarks that differ in the outcome being predicted should be scored within each of them before it is believed. The text-routing work reports per-dataset AUROC and warns explicitly about directions that capture dataset-specific cues [6]; we pooled, and pooling is what produced our headline. The check costs one line, and it was our own evaluation protocol rather than the method that failed it.

A Knowing *whether* is not knowing *where*

Our measurements show region choice matters more than action choice, so we tested whether the same internal signals locate the region. SmolVLM lays its 64 visual tokens in a contiguous 8×8 grid, so the states covering candidate cell k can be pooled,

$$z_k = \frac{1}{|\mathcal{P}_k|} \sum_{p \in \mathcal{P}_k} h_p^{(\ell)}, \quad (13)$$

and ranked by a probe trained on which cells actually answered correctly. This requires no judge model, no reinforcement learning and no extra forward pass.

On a 2×2 grid the localizer scores 48 to 51% across six depths against 51% for choosing uniformly at random, with a 64% oracle ceiling. Overlapping cells make random choice strong there, so we repeated the experiment at 3×3 and then at 4×4 . The finer grid is a fairer test: on the 400 examples all three share, the gap between random and the oracle widens from 12.2 to 16.2 points as cells shrink, so a localizer has more to find, not less.

One split is not enough to state a negative. Our earlier claim, that at no depth on either grid does the localizer beat random, rests on a single train/test split. Resampling it 60 times changes the sign of the answer (Table 24).

At 4×4 the localizer beats random at every depth, with intervals excluding zero. It also beats it by 0.9 accuracy points, which is 6% of the gap that a perfect localizer would close. **The honest claim is about magnitude, not sign:** the signal is real and it is too small to build on. We state it that way because we have insisted elsewhere

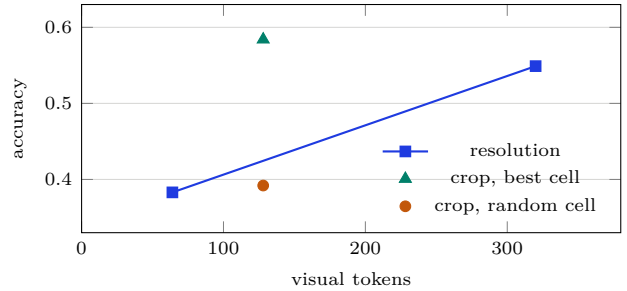


Figure 15: Two ways to spend a visual-token budget, $n = 1000$. The best crop sits above the resolution line at $2.5\times$ fewer tokens; a randomly chosen one sits on it. The vertical distance between the two crop markers is what a localizer would have to recover, and §A recovers 6% of it.

in this paper that a null is only a result when the interval is tight enough to have found an effect, and the converse obligation applies here.

What the localizer would have been worth. A negative result is easier to accept when the thing that failed is cheap. This one is not. Both ways of spending a visual-token budget are in the pilot, so they can be put on one axis (Table 25): a rung spends its tokens on the whole frame, a crop spends them on one cell at high magnification. This is also the comparison §3.2 has been making against the pruning literature [4] in words rather than numbers.

The two readings of the crop row carry the comparison (Fig. 15). A *randomly* chosen cell reaches 0.392, which is what the same tokens buy in resolution (0.383 at 64, and about 0.42 interpolated at 128). The *best* cell reaches 0.584, beating the dearest resolution configuration while spending $2.5\times$ fewer tokens. **Position is worth more per token than resolution, and only to a policy that knows where to look.**

That prices the failure. A working localizer would be the most token-efficient action in this action space, which is presumably why [3] is willing to spend cold-start fine-tuning, multi-turn GRPO and a 70B judge to obtain one. Our 0.9-point margin buys none of that headroom.

The low-resolution pass knows *whether* more resolution would help and essentially not *where* to spend it. This bounds what internal signals deliver at this scale, and suggests that the supervision machinery of [3] is not overengineering.

B What each part is worth

The components above are each justified where they are introduced, over five sections and on folds and cost models that were not always identical. That is how a paper acquires an inconsistency it cannot see, so we recompute all of them against one reference on one fold with one cost model.

Read at a fixed preference, an ablation says the opposite of the truth. Our first attempt held V constant and compared raw accuracy. Every removal appeared to *improve* accuracy, because every removal escalated more, and at fixed V nothing prevents a variant from buying accuracy with latency. That is the error this paper spends §5.1 warning about, committed on ourselves. Each variant is therefore swept to a frontier and read at a fixed latency budget (Table 26).

The losses are real but wildly unequal, so the table ranks what matters rather than listing what helps. **At 400 ms the ladder is what makes the budget usable at all:** binary escalation reaches 0.281, which is always-cheap to within 0.003, because no binary policy can afford the top rung inside that budget. And per-example pricing costs -0.003 $[-0.004, -0.001]$: an interval excluding zero, so a real effect, and small enough that we report it as negligible rather than as nothing. Those are different claims and the check that guards this table asserts the second.

C A second serving model, and cross-model transfer

Replication on a second serving model. Every result above has one model deciding for itself, so we repeated the entire chain (oracle run, activation extraction, probe fitting, rate sweep) with SmolVLM-256M as the model that answers, on the same 1000 images and questions (Table 27).

The finding replicates and strengthens. On the smaller model the probe’s advantage on the joint target widens from $+0.010$ to $+0.066$, and at the highest escalation rate both signals reach 0.455 accuracy while the probe costs 190.3 ms against 278.0 ms, 32% less, against 12 to 22% on the 500M.

Two properties survive the halving of model size and one does not. Self-knowledge is unchanged: AUROC on “will the cheap pass be correct” is 0.751 against 0.758, a difference of seven thousandths for a model with half the parameters and 16 fewer accuracy points. The escalation *opportunity* also grows, escalation repairs 24% of queries against 21%, and damages 3% against 4%. But knowing whether escalation will help degrades, from 0.734 to 0.672: half of the smaller model’s failures are unsolvable at any resolution, so the recoverability signal has less to work with. A weaker model needs routing more and can route slightly less well.

Cross-model transfer. Following the Encoder-Target Decoupling of [7], we predict the 500M model’s escalation outcomes from SmolVLM-256M activations on the same images. This scores 0.785 $[0.692, 0.873]$ at layer 30 and 0.770 at layer 1: at least as well as the 500M model predicts its own. The probe can therefore run off the serving path entirely, on a separate model that can be batched or placed on different hardware.

D An adversary on the allocation signal

The signal an attacker cannot see. A routing rule is an attack surface. Liu et al. [15] force deferral in multimodal cascades with a border-confined trigger that flattens the weak model’s output distribution, inflating the provider’s bill without changing any accuracy metric. Their attack is metric-agnostic precisely because it targets that distribution, which the probe never reads. As a cheap proxy we applied unoptimised border perturbations covering 36% of the image, with the probe fitted on training examples only (Table 28).

A perturbation requiring no optimisation, no model access and no knowledge of the deployed metric inflates compute by 24% relative when allocation reads output entropy, and by nothing when it reads the probe. We claim only the weak form: an attacker optimising the output distribution is not optimising a projection onto a direction they cannot observe, and here that indirection suffices. A white-box attacker who knows the probe direction is the case this does not settle.

E Abstention as a third action

We defined three targets in §3.3 and have used two. Q_3 (answerable at all?) is negative for 30.6% of the pilot: no configuration we run answers those queries correctly. A system that could recognise them would decline instead of spending, which is the third action the efficiency literature omits and the selective-prediction literature is built around [18, 19]. Two axes decide whether it is worth adding, and they disagree.

As a way to hold risk down, abstention is the weaker lever. Under a risk constraint the alternative to declining is to spend more and re-decide. Sweeping both thresholds and charging escalated queries for both passes (Table 29), escalation reaches coverage that abstention cannot.

At a 50% tolerance escalation reaches full coverage while escalating 38% of traffic. The comparison needs a cost-instrumented multi-configuration run, which is the artefact the efficiency literature builds and the selective-prediction literature does not, and it says the two fields have been pulling different levers on the same signal.

As a way to save compute, abstention is nearly worthless here, because the rule got there first. Escalating a query no configuration can answer is pure waste, and 30.0% of the test fold is in that state. If the gain rule of §7 spent its escalation budget blind to solvability, a perfect abstention gate would recover that entire share. It does not (Table 30).

Calibrating on the signed gain already suppresses these queries: they have $E[G \mid x] \approx 0$ whatever their error

probability, so the rule declines them for the same reason it declines confident failures. The residual headroom for a perfect, free abstention detector is 4.2% of latency at $V = 800$. **A third action is not where the remaining budget is.** What the escalation budget actually buys is visible in the same run: 54% of escalated queries have zero gain, escalation changes nothing, and that, not unanswerability, is the waste worth attacking.

If a guarantee is needed rather than a preference.

Both §7’s rule and the thresholds above are calibrated to a point estimate. Where an operator needs a distribution-free promise instead, the three-way split of [13] transfers directly, with its middle regime spending compute rather than widening a prediction set: at answer-level $\alpha = 0.5$ the policy answers 52%, escalates 38% and declines 9%, and its coverage guarantee holds out of sample to within $+0.025$. A recall floor in the style of [12] gives the complementary guarantee (retain at least 90% of the queries escalation would have fixed, achieved 97% on the test fold) and cannot collapse to never escalating by construction, which is the failure mode our cost-minimising tuners exhibited. We report these as available machinery rather than as our headline policy, since neither improves on the cost-derived rule when a preference can be stated.

References

- [1] S. Yang, J. Li, X. Lai, B. Yu, H. Zhao and J. Jia. Vision-Think: Smart and Efficient Vision Language Model via Reinforcement Learning. arXiv:2507.13348, 2025.
- [2] Z. Lin, Y. Liu, Y. Yang, L. Tao and D. Ye. AdaptVision: Efficient Vision-Language Models via Adaptive Visual Acquisition. arXiv:2512.03794v3, 2026.
- [3] N. Shabtay, M. Kimhi, A. Spector, S. Haray, E. Rivlin, C. Baskin, R. Giryas and E. Schwartz. Look Where It Matters: High-Resolution Crops Retrieval for Efficient VLMs. arXiv:2603.16932, 2026.
- [4] Q.-S. Zeng, Y. Li, Q. Wang, P.-T. Jiang, Z. Wu, M.-M. Cheng and Q. Hou. A Glimpse to Compress: Dynamic Visual Token Pruning for Large Vision-Language Models. arXiv:2508.01548, 2025.
- [5] W. Lugoloobi, T. Foster, W. Bankes and C. Russell. LLMs Encode Their Failures: Predicting Success from Pre-Generation Activations. ICLR 2026 LIT Workshop; arXiv:2602.09924v3, 2026.
- [6] I. V. Moreno Cencerrado, A. Gonzalez Hawthorne et al. No Answer Needed: Predicting LLM Answer Accuracy from Question-Only Linear Probes. ICLR 2026 Workshop on Principled Design for Trustworthy AI; arXiv:2509.10625v3, 2026.
- [7] T. Varshney, A. Surla, M. Xu, G. V. Krishnan, M. Jebllick, D. Austin, N. Vaidya and D. Onofrio (NVIDIA). LLM Router: Rethinking Routing with Prefill Activations. arXiv:2603.20895v2, 2026.
- [8] J. Zhan, H. Shen and M. Guo et al. Seeing is Free, Speaking is Not: Uncovering the True Energy Bottleneck in Edge VLM Inference. arXiv:2607.09520, 2026.
- [9] A. Marafioti, O. Zohar, M. Farré, M. Noyan, E. Bakouch et al. (Hugging Face). SmolVLM: Redefining small and efficient multimodal models. arXiv:2504.05299, 2025.
- [10] H. Shin, C. Kim, R. Kim, H. Yoo and J. Kim. Rethinking Small VLM Quantization: From Component-Wise Analysis to Hardware-Aware Edge Deployment. arXiv:2607.08029, 2026.
- [11] V. Kotte. UCCI: Calibrated Uncertainty for Cost-Optimal LLM Cascade Routing. arXiv:2605.18796, 2026.
- [12] K. Ruan, Z. Huang, Z. Zhou, Q. Wei, J. Lin, X. Wang and H. Sun. Doomed from the Start: Early Abort of LLM Agent Episodes via a Recall-Controlled Probe Cascade. arXiv:2607.06503v2, 2026.
- [13] S. Tayebati, D. Kumar, N. Darabi, D. Jayasuriya, R. Krishnan and A. R. Trivedi. Learning Conformal Abstention Policies for Adaptive Risk Management in Large Language and Vision-Language Models. arXiv:2502.06884, 2025.
- [14] Y. Moslem, M. Kacmajor, V. Nedumpozhimana, A. Abbas, S. Panahi et al. Cluster, Route, Escalate: Cascaded Framework for Cost-Aware LLM Serving. arXiv:2606.27457, 2026.
- [15] Z. Liu, Y. Zeng, Y. Chang and L. Lin. Forced Deferral: Manipulating Routing Decisions in Multimodal LLM Cascades. arXiv:2606.15308, 2026.
- [16] Z. Sun, D. Chen and S. Li. When Efficiency Backfires: Cascading LLMs Trigger Cascade Failure under Adversarial Attack. arXiv:2605.17288, 2026.
- [17] S. Qin, H. Yu, C. Wu, Z. Li et al. (Qwen Team, Alibaba). VLMCache: Computing 2% Vision Tokens and Reusing 98% for Vision-Language Inference. arXiv:2512.12977v2, 2025.
- [18] N. Madhusudhan, V. Yadav and A. Lacoste (ServiceNow Research). Knowing When Not to Answer: Evaluating Abstention in Multimodal Reasoning Systems. arXiv:2604.14799, 2026.
- [19] T. Srinivasan, J. Hessel, T. Gupta, B. Y. Lin, Y. Choi, J. Thomason and K. R. Chandu. Selective “Selective Prediction”: Reducing Unnecessary Abstention in Vision-Language Reasoning. arXiv:2402.15610, 2024.

claim	status
The rate parameter is removable given a stated V (Prop. 1)	demonstrated analytically, and it is a transposition of [11]
Entropy does not clear the randomisation hull	measured on one mixture, both costings, all operating points
A free image descriptor matches the probe	measured on one mixture; untested on traffic where size and content decouple
The two regimes require opposite signals	the free descriptor fails on both workloads measured; the graded ladder clears on the one where the model is strong, marginally on the other
The probe’s pooled advantage is a domain effect	measured three ways; the magnitude within a domain is uncertain
Accuracy saturates by 1152 px	measured on two document corpora and five models; untested outside document QA
Pixels bind and tokens do not	measured on the one model whose two axes move separately; not decomposed for the other four
The ceiling bounds any model’s need from above	observed on five models, one counterexample would refute it
The ladder pays where rungs are distinct	measured on DocVQA; one preference of four on InfographicVQA; null where no top rung exists
Escalation value is domain-determined	observed across four datasets, assumed to extend to unseen ones

Table 23: Evidential status of the paper’s claims. “Measured” means an interval was computed on held-out data; “observed” means the pattern is consistent across the runs we have without an interval that would detect its absence.

	2×2 ($n = 500$)	4×4 ($n = 400$)
depths beating random	1 of 6	4 of 4
best margin	+0.006	+0.009 [0.005, 0.014]
headroom to the oracle	0.122	0.162
share of the gap closed	n/a	6%

Table 24: Localizer margin over random cell choice, 60 resampled splits. Random is recomputed on each test fold rather than assumed, since the cell-correctness rate varies by fold.

family	configuration	tokens	accuracy
resolution	384 px	64	0.383
resolution	768 px	320	0.516
resolution	full	320	0.549
crop, random cell	2×2	128	0.392
crop, best cell	2×2	128	0.584

Table 25: Accuracy against visual tokens, $n = 1000$, two ways of spending the same budget. Token counts are medians: the 768 px and full configurations share one because the processor caps its target at the input’s longest side, and they differ on the 44% of images large enough for the cap not to bind (§7.2). Paired against the best resolution configuration, the best crop is +0.035 [0.008, 0.063] more accurate at 2.5× fewer tokens.

variant	acc. at 400 ms	acc. at 500 ms
reference	0.653	0.743
no per-example pricing	0.650 (−0.003)	0.744 (+0.0004)
no calibration (tuned rate)	0.431 (−0.221)	0.571 (−0.172)
no ladder (binary)	0.281 (−0.372)	0.622 (−0.121)
no gain target (UCCI)	0.278 (−0.375)	0.611 (−0.133)
no signal (fixed policy)	0.278 (−0.375)	0.278 (−0.465)

Table 26: Each variant swept over six operator preferences and read at a fixed budget, 1200 DocVQA pages, 30 resamples. Deltas are paired against the reference frontier.

serving model	cheap acc.	full acc.	probe AUROC	entropy AUROC
SmolVLM-500M	0.383	0.549	0.761	0.751
SmolVLM-256M	0.298	0.511	0.682	0.616

Table 27: AUROC is on the joint escalation-gain target. On the reported fold of both models every entropy operating point is Pareto-dominated; six probe points lie on the front for the 256M model, four for the 500M. The resplit test of Fig. 7 was run for the 500M model only.

perturbation	signal	shift	escalation
high-frequency noise	entropy	+0.28 σ	50% → 62%
high-frequency noise	probe	−0.04 σ	50% → 50%
flat fill	entropy	+0.19 σ	50% → 56%
flat fill	probe	+0.01 σ	50% → 50%

Table 28: Border perturbations touching no central pixel, 80 held-out images. Two structurally opposite perturbations both raise output entropy and neither moves the probe.

risk tolerance	abstain only	or escalate	cost
$\leq 30\%$	unreachable	58%	320.8
$\leq 40\%$	41%	88%	312.2
$\leq 50\%$	69%	100%	200.7

Table 29: Maximum coverage at each risk tolerance, latency in milliseconds. At a 30% tolerance abstention alone serves no query at acceptable risk.

V	escalates	of which hopeless	perfect gate saves
400	24%	17%	2.8%
800	34%	19%	4.2%
1600	40%	22%	5.3%

Table 30: Escalation is aimed below the 30.0% base rate of unsolvable queries, so an oracle abstention gate (which cannot lose accuracy, since those queries are wrong under every action) recovers only the margin.